Hierarchical Progressive Transform With Learnable Prior for Hyperspectral Image Super-Resolution

Zhihui Tu, Yisha Gao, Wenyu Hu, Yong Chen, Honghui Xu, *Member, IEEE*, and Jian Lu

***Abstract*—Hyperspectral image super-resolution (HSI-SR) aims to reconstruct a high-resolution HSI (HR-HSI) by fusing a low-resolution HSI (LR-HSI) with an HR multispectral image (HR-MSI). Tensor nuclear norm (TNN) regularization exploits transform domains to capture the low-rank structure of HSIs. However, existing methods face a fundamental rigidity–flexibility tradeoff: fixed transforms offer stability but lack adaptability, while data-driven transforms provide flexibility but suffer from optimization instability. To address this, we propose a novel HSI-SR model based on a Log-sum-based full-scale learnable TNN (LLTNN) regularizer. The core of our framework is the hierarchical progressive transform (HPT) operator, which establishes a principled dual-layer scheme. The explicit layer employs fixed multidomain transforms to provide a robust structural anchor, while the implicit layer introduces a data-driven adaptive transform that captures scene-specific correlations that fixed bases cannot model. This explicit-to-implicit design yields more compact low-rank representations than single-domain or simple cascade methods. The full-scale framework imposes low-rankness jointly across all three tensor modes with a nonconvex log-sum penalty to preserve essential features while suppressing noise. Extensive experiments on simulated and real-world datasets demonstrate that our method achieves state-of-the-art performance compared with existing approaches. The code is available at https://github.com/Tuzhihui/LLTNN_code**

***Index Terms*—Hierarchical progressive transform (HPT), hyperspectral image super-resolution (HIS-SR), image fusion, multidomain, tensor nuclear norm (TNN).**

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I. Introduction

HYPERSPECTRAL image (HSI), an advanced sensing modality, enables synergistic perception of spatial morphology and spectral characteristics by capturing hundreds of contiguous narrow bands [1]. This fine-grained material distinction facilitates significant potential in diverse fields, such as land-cover classification [2] and change detection [3]. However, HSI systems inherently face a fundamental tradeoff between spatial and spectral resolution due to the physical constraints of imaging sensors. Consequently, sensors typically capture either a low-resolution HSI (LR-HSI), which is spectrally rich but spatially coarse, or a high-resolution multispectral image (HR-MSI), which is spatially sharp but spectrally sparse [4], [5]. To overcome this hardware bottleneck, which severely limits fine-grained scene analysis, hyperspectral image super-resolution (HSI-SR) has emerged as a crucial computational technique, aiming to reconstruct an HR-HSI by fusing the complementary information from LR-HSI and HR-MSI [6], [7], [8].

Within the context of fusion-based HSI-SR, current methodologies primarily diverge into two paradigms: data-driven *deep learning (DL)* approaches and traditional *model-based* approaches. While DL-based methods have achieved impressive results [9], [10], these data-driven paradigms possess inherent limitations. Their performance depends heavily on large-scale and well-aligned training datasets—resources that are often impractical or even impossible to obtain in real-world remote-sensing scenarios. Furthermore, the black-box nature of most deep networks lacks clear physical interpretability, thereby limiting their trustworthiness in critical applications [11], [12], [13]. Faced with these challenges, we shift our focus to model-based strategies, which offer stronger interpretability and model robustness while avoiding reliance on large-scale training data [14], [15], [16]. These methods formulate HSI-SR as a mathematical optimization problem, typically governed by the imaging observation model and constrained by regularization terms that encode the intrinsic structural properties of the target HR-HSI.

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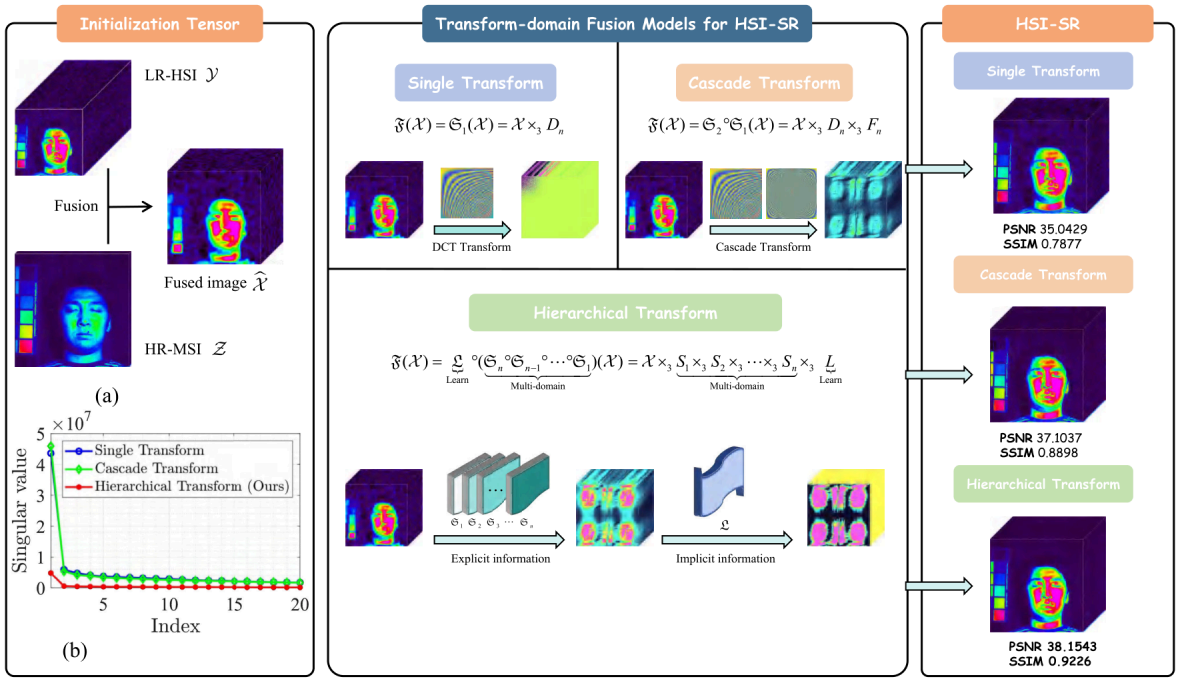


Fig. 1. Motivating comparison of different transform strategies within the transform-domain HSI-SR framework on the face dataset. Here, the "fused image" is the in-loop estimate of the target HR-HSI, initialized by the first closed-form Sylvester solve of the coefficient subproblem (with auxiliary variables and multipliers set to zero) and iteratively refined by the proposed framework, rather than the output of any external fusion algorithm. (a) Initialization. (b) Singular value curve.

Among model-based priors, low-rank tensor formulations, particularly those built upon tensor singular value decomposition (t-SVD), have emerged as a dominant strategy. However, the efficacy of the tensor nuclear norm (TNN) framework is fundamentally governed by the transform operator used to expose the latent low-rank structure. Existing methods generally face a critical *rigidity–flexibility tradeoff* in selecting this transform.

Most traditional methods [17], [18], [19] rely on generic-prior transformations, such as the fixed discrete Fourier transform (DFT) or discrete cosine transform (DCT). These approaches leverage the global smoothness prior of HSIs, offering desirable spectral-domain interpretability and computational stability. However, such data-agnostic transforms presume a fixed frequency decay pattern (a rigid view), which often fails to capture the complex, nonstationary statistical characteristics of varying scenes, resulting in inefficient singular value concentration for scene-specific structures. As shown by the blue curve in Fig. 1(b), the slow decay of singular values under a single DCT (PSNR = 35.04 dB) indicates insufficient information concentration.

Conversely, methods utilizing instance-specific transformations, such as those based on pure dictionary learning or self-adaptive learning [20], offer high adaptivity. They can theoretically capture implicit statistical structures that fixed bases miss. Yet, without the regularization of a stable basis, these fully flexible methods face vast optimization spaces and are prone to noise sensitivity or local optima—an overfitting or instability risk. While recent advanced attempts have sought to extend beyond single domains, for instance, [21] introduces successive linear transforms to construct a cross-domain representation, and [22] explores cascading DFT with self-learning matrices—they typically treat these combinations as a simple serial concatenation (e.g., mathematically chaining two operators). They lack a mechanism to systematically decouple the roles of fixed and adaptive components, i.e., to clarify what each transform layer should capture. Consequently, as evidenced by the green curve in Fig. 1(b), the cascade of DCT and DFT achieves only moderate improvement (PSNR = 37.10 dB, +2.1 dB over single DCT) compared to our hierarchical strategy.

In the context of remote sensing, this rigidity–flexibility tradeoff becomes particularly prominent and detrimental due to the unique physical characteristics of real-world land-cover distributions. Specifically, remote sensing scenes typically exhibit a complex duality. On the macrolevel, they contain large homogeneous regions (e.g., extensive water bodies, forests, or agricultural fields) that require highly stable, global structural priors to prevent spectral distortion. On the micro level, however, these scenes suffer from severe spatial heterogeneity, presenting nonstationary local textures, intricate urban structures, and mixed pixels at class boundaries. A purely rigid transform effectively models the global smoothness of homogeneous areas but inevitably oversmooths critical high-frequency heterogeneous details. Conversely, fully flexible, data-driven transforms are prone to overfitting and noise amplification, especially within those broad homogeneous zones. Furthermore, unlike natural images, remote sensing HSIs possess a strong physically coupled spatio-spectral structure; treating them with single-mode regularization often violates the intrinsic physical consistency of material reflectance. Therefore, it is highly imperative to design a framework tailored to remote sensing demands. To this end, we introduce the dual-layer design in *hierarchical progressive transform*

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(HPT) to simultaneously anchor macro-homogeneous structures and adaptively capture micro-heterogeneous details, while employing the full-mode regularization in *log-sum-based full-scale learnable TNN (LLTNN)* to strictly preserve the global spatiospectral coupling.

To resolve this dilemma and achieve a synergistic unification of robustness and flexibility, we propose a novel HSI-SR framework driven by an HPT strategy. The core innovation of HPT lies not in any individual transform, but in the principled decoupling mechanism that assigns distinct and nonoverlapping roles to each layer. Unlike prior cascade methods that simply chain transforms without role differentiation, HPT establishes a coarse-to-fine refinement pipeline where each layer operates on a progressively constrained subspace.

1) *Layer 1 (Explicit Transform):* This layer provides the structural foundation of HPT by deploying a carefully designed multidomain transform that cascades complementary bases (e.g., DFT followed by DCT). In remote sensing contexts, this design targets the predictable, cross-scene regularities: DFT exploits the inter-band correlations arising from atmospheric absorption windows and material reflectance physics, while DCT compacts energy for spatially smooth regions, such as homogeneous water bodies or vegetation canopies. The key contribution of this layer is not the individual transforms (which are well-known), but their synergistic combination that maximally compresses the effective search space for the subsequent adaptive layer.
2) *Layer 2 (Implicit Transform):* Building upon the compressed representation from Layer 1, this layer deploys a data-driven adaptive transform to capture the residual structures that fixed bases cannot model. The term *implicit* indicates that the transform bases emerge from data rather than being prescribed. Crucially, the implicit layer does not operate on the raw tensor, but on the compacted latent space left by the explicit layer. This constrained learning setting is what distinguishes HPT from purely data-driven methods: the search space is preconditioned by Layer 1, preventing overfitting while retaining adaptability.

This hierarchical design allows us to progressively concentrate HSI information from general structural patterns to scene-specific details. As validated in Fig. 1(b), the red curve corresponding to our HPT strategy (DCT+DFT+Learnable) achieves the sharpest singular value decay and the highest reconstruction accuracy (PSNR = 38.15 dB, +3.1 dB over single DCT), demonstrating the most compact low-rank representation among all compared strategies. Building upon this, we propose the LLTNN regularizer, which imposes this hierarchical low-rankness jointly across all three tensor modes to capture global spatiospectral couplings.

In summary, the main contributions of this article are given as follows.

1) We propose the HPT operator, whose core innovation is a principled decoupling mechanism that assigns nonoverlapping roles to two hierarchical layers. HPT ensures that the explicit layer captures global structures, while the implicit layer exclusively refines residuals within a constrained subspace. This design simultaneously achieves the stability of fixed transforms and the adaptability of learned transforms.
2) We formulate a multidomain transform-based TNN framework within the t-SVD paradigm and instantiate the associated tensor algebraic operations, including a generalized t-product, t-SVD, and the corresponding TNN.
3) We derive an efficient optimization algorithm based on the alternating direction method of multipliers (ADMM) to solve the resulting minimization problem. Extensive experiments on both simulated and real-world datasets demonstrate that the proposed LLTNN method achieves state-of-the-art performance.

The overall process framework of the proposed method is shown in Fig. 2. The remainder of this article is organized as follows. Section II reviews related work. Section III introduces notations and preliminaries. Section IV presents the proposed HPT operator and LLTNN regularizer. Section V derives the ADMM-based optimization algorithm. Section VI reports experimental results. Section VII concludes this article.

II. Related Work

This section reviews the research landscape of HSI-SR. We first briefly discuss data-driven DL approaches and then focus on the evolution of tensor-based methods, tracing the trajectory from single-transform paradigms to the recent hybrid strategies that directly inspire our work.

A. DL-Based Approaches

In recent years, DL methods have become a prominent research direction in HSI-SR due to their powerful feature extraction and nonlinear fitting capabilities. In particular, methods based on convolutional neural networks (CNNs) [4], [5], [23], [24], Transformer models [9], [25], [26], [27], and diffusion models [10], [28] have been widely applied with significant advancements. For instance, Yuan et al. [23] designed a multiscale and multidepth CNN (MSDCNN) that incorporates multiscale feature extraction and residual learning into the CNN architecture. Wang et al. [25] proposed a novel multihierarchical cross-transformer (MCT-Net), which performs cross-modal interaction and feature reconstruction on multiscale features extracted from HSI and MSI. Subsequently, some methods have approached the problem from the perspective of pixel-aware refinement [29], [30], [31], leveraging image-agnostic and image-specific characteristics to learn image priors and enhance model generalization. More recently, model-inspired deep methods have also been explored for related multiband fusion tasks, such as pan-sharpening [32]. To alleviate the dependence on labeled training data, several recent unsupervised DL approaches have been proposed. Liu et al. [33] designed a model-inspired autoencoder (MIAE) that integrates NMF-based spectral decomposition into an encoder–decoder architecture, enabling unsupervised fusion without paired training samples. Subsequently, Liu et al. [34] proposed a spectral diffusion prior (SDP) that leverages a

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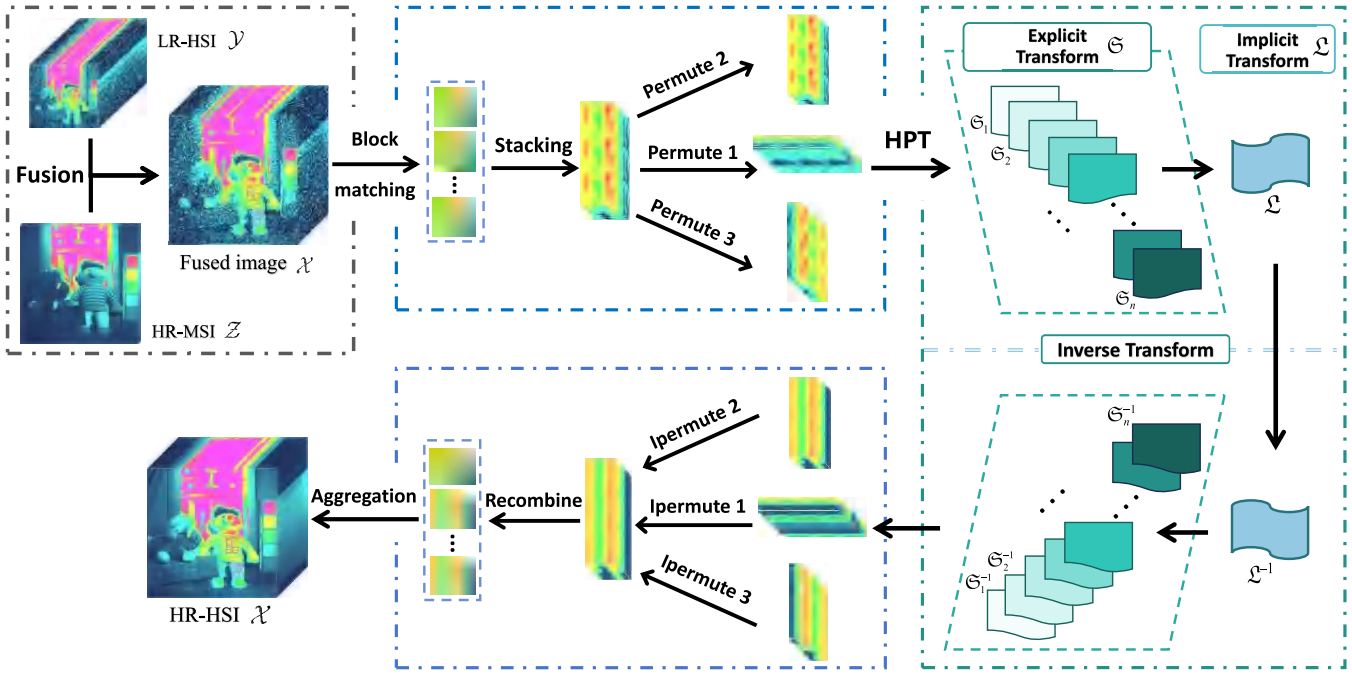


Fig. 2. Overall flowchart of the proposed LLTNN model architecture. As in Fig. 1, the "fused image" here refers to the in-loop estimate of the HR-HSI, obtained from the first closed-form Sylvester solve of the coefficient subproblem and iteratively updated by the proposed algorithm in subsequent ADMM iterations.

pretrained diffusion model to capture the spectral distribution of HSIs, further advancing the unsupervised paradigm.

Despite these efforts, unsupervised DL methods still require careful architectural design and rely on implicitly learned priors whose physical meaning is not always transparent. Faced with these challenges, we turn our attention to model-based methods that offer stronger interpretability and robustness. Within this paradigm, early efforts primarily adopt matrix factorization strategies, which decompose the target HR-HSI into spectral and spatial factors and impose priors, such as sparsity or smoothness on these factors. Representative methods include coupled nonnegative matrix factorization (CNMF) [35], nonnegative structured sparse representation (NSSR) [36], and subspace-based approaches with various regularizations [37]. While effective, these formulations unfold the 3-D HSI into 2-D matrices, inevitably disrupting the native multidimensional correlations among spatial and spectral modes. To preserve the intrinsic tensor structure, recent works have shifted toward tensor decomposition frameworks, particularly the t-SVD and its associated TNN, which we review in the following.

B. Single Transform-Based Tensor Methods

Model-based tensor methods primarily rely on the t-SVD framework, where the choice of the transform operator dictates the nature of the low-rank prior. Historically, these methods have operated within a *single* transform domain, evolving from rigid to adaptive definitions.

Early works utilized fixed, data-agnostic transforms. The seminal TNN employed the DFT to reveal low-rank structures in the frequency domain. Following this, variants based on DCT and Framelet transform were proposed to capture specific physical properties, such as energy compaction or multiscale features. While physically interpretable and computationally efficient, these rigid priors impose a universal assumption on all images, failing to adapt to nonstationary statistical characteristics.

To overcome this rigidity, researchers introduced data-driven, learnable transforms. Methods, such as SALTS [20] and LS2T2NN [38], learn an adaptive transform matrix directly from the data, offering high flexibility. However, without the regularization of a stable basis, these purely data-driven methods often lack a global structural anchor. As a result, both categories share a common limitation: they rely on a single view of the data and, thus, cannot simultaneously exploit the stability offered by explicit physical priors and the flexibility enabled by implicit statistical adaptation.

C. Hybrid Transform-Based Methods

Recognizing the limitations of single-domain transforms, recent state-of-the-art methods have begun to explore multidomain or cascade strategies, which serve as the immediate predecessors to our work.

CroDoSR [21] introduced the concept of cross-domain rank, utilizing successional linear transforms (e.g., Fourier followed by Framelet) to progressively sparsify the tensor. Similarly, Xu et al. [22] proposed the cascade-transform-based TNN (CTNN), investigating hybrid variants, such as appending a self-learning matrix to the DFT. These methods represent a significant step forward, demonstrating that combining transforms leads to better singular value concentration than single-domain approaches.

Beyond linear cascades, recent works have advanced the tensor transform paradigm along two directions. Along the *nonlinear transform* direction, Zhang et al. [39] and Xu et al. [40] introduced learnable nonlinear mappings into the tensor

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low-rank framework, and most recently, Xu et al. [41] developed a nonlinear triple-domain transform TNN for HSI-SR, applying elementwise nonlinear activations between cascaded linear layers. Along the *spatial–spectral structural prior* direction, Peng et al. [14] and Ma et al. [42] designed mode-specific low-rank and sparsity constraints tailored to the joint spatial–spectral correlations of HSIs, while Xue et al. [37], [43], [44], [45] explored parametric multilayer tensor sparsity and spatial–spectral structured representations for high-dimensional image recovery.

Despite these advances, existing methods, whether cascade, nonlinear, or spatial–spectral, do not explicitly decouple the distinct roles of fixed and adaptive components within the transform design: nonlinear methods interleave activations without clarifying what each layer should capture, and spatial–spectral methods design specific mode priors without a unified hierarchical progression. In contrast, our HPT framework assigns distinct and complementary roles to each layer: the explicit layer leverages a multidomain design to capture correlations from multiple perspectives, while the implicit layer focuses exclusively on the residual structures that fixed bases cannot model.

III. Notations and Preliminaries

A. Notations

This section briefly introduces the notations and preliminary definitions used throughout this article, laying the foundation for subsequent theoretical derivations and model formulations. We use x, $\mathbf{x}$, $\mathbf{X}$, and $\mathcal{X}$ to denote scalars, vectors, matrices, and tensors, respectively, in this article. For an order-3 tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, I_k denotes the size of the kth mode of $\mathcal{X}$ for $k = 1, 2, 3$. Its (a, b, c)th element is denoted as $\mathcal{X}(a, b, c)$ or x_{abc}, the qth frontal slice is denoted as $\mathcal{X}(:, :, q)$ or $\mathcal{X}^{(q)}$, and its mode-k matricization is expressed as $\mathbf{X}_{(k)} \in \mathbb{R}^{I_k \times \prod_{j \neq k} I_j}$. The block diagonal matrix of $\mathcal{X}$, denoted as $\text{bdiag}(\mathcal{X}) \in \mathbb{R}^{I_1 I_3 \times I_2 I_3}$, is given by

$$\text{bdiag}(\mathcal{X}) = \begin{bmatrix} \mathcal{X}^{(1)} & & & \\ & \mathcal{X}^{(2)} & & \\ & & \ddots & \\ & & & \mathcal{X}^{(I_3)} \end{bmatrix}$$

and the block circulant matrix of $\mathcal{X}$, denoted as $\text{bcirc}(\mathcal{X})$, is defined as

$$\text{bcirc}(\mathcal{X}) = \begin{bmatrix} \mathcal{X}^{(1)} & \mathcal{X}^{(I_3)} & \cdots & \mathcal{X}^{(2)} \\ \mathcal{X}^{(2)} & \mathcal{X}^{(1)} & \cdots & \mathcal{X}^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathcal{X}^{(I_3)} & \mathcal{X}^{(I_3-1)} & \cdots & \mathcal{X}^{(1)} \end{bmatrix}.$$

The Frobenius norm is denoted as $\|\mathcal{X}\|_F = \left(\sum_{ijk} x_{ijk}^2\right)^{1/2}$. The matrix nuclear norm of $\mathbf{X}$ is $\|\mathbf{X}\|_* = \sum_i \sigma_i(\mathbf{X})$, where $\sigma_i(\mathbf{X})$ is the ith singular value of $\mathbf{X}$, and the spectral norm is $\|\mathbf{X}\|_\sigma = \max_i \sigma_i(\mathbf{X})$.

B. Preliminaries on t-SVD Framework

We briefly review the classic t-SVD framework based on the DFT, which serves as the baseline for our proposed method.

Definition 1 (Tensor Mode-k Product [46]): For any third-order $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ and a matrix $\mathbf{Y} \in \mathbb{R}^{J_k \times I_k}$, tensor mode-$k$ product is defined as

$$\mathcal{Z} = \mathcal{X} \times_k \mathbf{Y} = \text{fold}_k\left(\mathbf{Y}\mathbf{X}_{(k)}\right) \tag{1}$$

where $\mathbf{X}_{(k)}$ is the mode-k matricization of $\mathcal{X}$ and $\text{fold}_k(\cdot)$ is the corresponding inverse operator of matricization that rearranges elements of a matrix into a tensor.

Definition 2 (t-Product [46]): For tensors $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ and $\mathcal{Y} \in \mathbb{R}^{I_2 \times I_4 \times I_3}$, the t-product $\mathcal{Z} = \mathcal{X} * \mathcal{Y} \in \mathbb{R}^{I_1 \times I_4 \times I_3}$ is computed via

$$\overline{\mathcal{Z}} = \overline{\mathcal{X}} \cdot \overline{\mathcal{Y}} \tag{2}$$

where $\overline{\mathcal{X}} = \text{fft}(\mathcal{X}, [], 3)$ denotes the Fourier transform along the third mode and $\cdot$ is block-diagonal matrix multiplication. The final result in the original domain is obtained via the inverse FFT

$$\mathcal{Z} = \text{ifft}\left(\overline{\mathcal{Z}}, [], 3\right). \tag{3}$$

Definition 3 (t-SVD [46]): Any tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ can be factorized as

$$\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T \tag{4}$$

where $\mathcal{U} \in \mathbb{R}^{I_1 \times I_1 \times I_3}$ and $\mathcal{V} \in \mathbb{R}^{I_2 \times I_2 \times I_3}$ are orthogonal tensors and $\mathcal{S} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ is f-diagonal (each frontal slice is diagonal).

Definition 4 (TNN [47]): The TNN of $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ is defined as

$$\|\mathcal{X}\|_{\text{TNN}} = \sum_{i=1}^{I_3} \left\|\overline{\mathcal{X}}^{(i)}\right\|_* \tag{5}$$

where $\overline{\mathcal{X}}^{(i)}$ is the ith frontal slice of $\overline{\mathcal{X}} = \text{fft}(\mathcal{X}, [], 3)$ and $\|\cdot\|_*$ is the matrix nuclear norm.

C. HSI-SR Problem Formulation

HSI-SR aims to reconstruct an HR-HSI $\mathcal{X} \in \mathbb{R}^{H \times W \times S}$ with both fine spatial details and rich spectral information, by fusing an LR-HSI $\mathcal{Y} \in \mathbb{R}^{h \times w \times S}$ and an HR-MSI $\mathcal{Z} \in \mathbb{R}^{H \times W \times s}$, where $h \ll H$, $w \ll W$, and $s \ll S$ of the same scene. Mathematically, $\mathcal{Y}$ can be modeled as a spatially degraded version of the underlying HR-HSI $\mathcal{X}$, and $\mathcal{Z}$ is a spectrally degraded version of $\mathcal{X}$. For the mode-3 matricization of tensors, this degradation process is expressed as

$$\mathbf{Y}_{(3)} = \mathbf{X}_{(3)}\mathbf{B}\mathbf{S}, \quad \mathbf{Z}_{(3)} = \mathbf{R}\mathbf{X}_{(3)} \tag{6}$$

where $\mathbf{Y}_{(3)} \in \mathbb{R}^{S \times hw}$, $\mathbf{Z}_{(3)} \in \mathbb{R}^{s \times HW}$ and $\mathbf{X}_{(3)} \in \mathbb{R}^{S \times HW}$ are the mode-3 unfolded matrices of $\mathcal{Y}$, $\mathcal{Z}$, and $\mathcal{X}$, respectively. $\mathbf{B} \in \mathbb{R}^{HW \times HW}$ denotes the spatial blurring matrix, $\mathbf{R} \in \mathbb{R}^{s \times S}$ is the spectral response matrix, and $\mathbf{S} \in \mathbb{R}^{HW \times hw}$ represents the spatial downsampling matrix, which selects corresponding pixels to reduce spatial resolution.

Based on the above degradation models, the primary goal of HSI-SR is to estimate $\mathcal{X}$ by minimizing the data fidelity loss between the reconstructed and observed images, i.e.,

$$\min_{\mathcal{X}} \|\mathbf{Y}_{(3)} - \mathbf{X}_{(3)}\mathbf{B}\mathbf{S}\|_F^2 + \|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{X}_{(3)}\|_F^2. \tag{7}$$

Simultaneously, since the spectral vectors of an HSI typically lie within a low-dimensional subspace, the target HR-HSI $\mathcal{X}$

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can be decomposed into a spectral basis $\mathbf{D} \in \mathbb{R}^{S\times L}$ (where $L \ll S$) and a coefficient tensor $\mathcal{A} \in \mathbb{R}^{H\times W\times L}$

$$\mathbf{X}_{(3)} = \mathbf{D}\mathbf{A}_{(3)}. \tag{8}$$

By substituting (8) into the observation models (7), the HSI-SR task is effectively reformulated as solving for the coefficient tensor $\mathcal{A}$

$$\min_{\mathcal{A}} \|\mathbf{Y}_{(3)} - \mathbf{D}\mathbf{A}_{(3)}\mathbf{B}\mathbf{S}\|_F^2 + \|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{D}\mathbf{A}_{(3)}\|_F^2. \tag{9}$$

However, (9) represents an ill-posed inverse problem, and relying solely on data fidelity is insufficient to recover the fine details of $\mathcal{A}$. While standard low-rank regularizers (e.g., TNN) have been widely used, they are limited by fixed transform bases (like DFT) and fail to capture the nonlocal self-similarity inherent in HSIs. To address these limitations, we propose a novel regularization framework in Section IV. The key idea is to design an HPT operator that systematically decouples fixed multidomain bases from data-driven adaptive components, combined with a nonlocal patch-based regularization strategy.

IV. Proposed Model

Before delving into mathematical details, we provide an intuitive overview of the complete processing pipeline (see Fig. 2). Given an observed LR-HSI $\mathcal{Y}$ and HR-MSI $\mathcal{Z}$, our method reconstructs the target HR-HSI $\mathcal{X}$ through four interconnected stages.

1) *Spectral Subspace Decomposition (see Section III-C):* To reduce computational burden and exploit high spectral correlation, $\mathcal{X}$ is factorized as $\mathbf{X}_{(3)} = \mathbf{D}\mathbf{A}_{(3)}$, simplifying the task to estimating the low-dimensional coefficient tensor $\mathcal{A} \in \mathbb{R}^{H\times W\times L}$.
2) *Nonlocal Patch Grouping (see Section IV-B):* Guided by the HR-MSI's spatial structure, similar patches of $\mathcal{A}$ are clustered into 3-D tensor groups $\{\mathcal{A}^j\}_{j=1}^{C}$, transforming spatial redundancy into a highly low-rank format.
3) *HPT and LLTNN Regularization (see Section IV-A):* Each patch group undergoes the HPT. Its explicit layer anchors global structural priors, while the implicit layer (data-driven SVD) adaptively captures scene-specific residuals. The compact representation is then constrained by the LLTNN regularizer across all three tensor modes to preserve essential features while suppressing noise.
4) *ADMM Optimization (see Section V):* The nonconvex objective is efficiently solved via an ADMM framework. It alternates between a Sylvester-based coefficient update and log-sum tensor singular value thresholding (LTSVT)-based regularization updates, with the implicit-layer basis updated at every iteration.

The remainder of this section formalizes stages 2 and 3. Section IV-A derives the HPT operator and LLTNN regularizer, while Section IV-B presents the nonlocal patch-based formulation.

A. LLTNN

As established in Section I, existing transform-based methods face a fundamental rigidity–flexibility tradeoff. To address this, we propose the HPT, a principled dual-layer scheme that decouples explicit (fixed multidomain) and implicit (data-driven and adaptive) components. This section formalizes HPT and the resulting LLTNN regularizer.

Recall that HPT consists of two layers: 1) an *explicit* layer comprising fixed multidomain transforms and 2) an *implicit* layer comprising a data-driven adaptive transform. The following definition provides the precise mathematical formulation.

Definition 5 (HPT): Let $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times K_0}$ be a third-order tensor, where K_0 denotes the original third-mode dimension. The HPT $\mathfrak{F}(\mathcal{X})$ is a composite transform consisting of two layers.

1) *Explicit Layer* $\mathfrak{S}$: A cascade of n predefined linear transforms (e.g., DFT and DCT) applied directly to the input tensor $\mathcal{X}$, providing a fixed multidomain basis. Its output $\mathcal{Y} = \mathfrak{S}(\mathcal{X})$ explicitly serves as the preconditioned input for the subsequent adaptive layer $\mathfrak{L}$.
2) *Implicit Layer* $\mathfrak{L}$: A data-driven adaptive transform learned via SVD, capturing scene-specific structures.

Its mathematical form is defined as

$$\mathfrak{F}(\mathcal{X}) = \mathfrak{L} \circ \mathfrak{S}(\mathcal{X}) \tag{10}$$

where the successional linear transform $\mathfrak{S}$ cascades n fixed linear transforms with transformation matrices $\mathbf{S}_i \in \mathbb{R}^{K_i\times K_{i-1}}$, $i = 1, \ldots, n$. Specifically,

$$\begin{aligned}\mathfrak{S}(\mathcal{X}) &= (\mathfrak{S}_n \circ \mathfrak{S}_{n-1} \circ \cdots \circ \mathfrak{S}_1)(\mathcal{X}) \\ &= \mathcal{X} \times_3 \mathbf{S}_1 \times_3 \mathbf{S}_2 \times_3 \cdots \times_3 \mathbf{S}_n.\end{aligned} \tag{11}$$

Driven by data, the transform $\mathfrak{L}$ uncovers the latent structure within $\mathfrak{S}(\mathcal{X})$. Let $\mathcal{Y} = \mathfrak{S}(\mathcal{X}) \in \mathbb{R}^{I_1\times I_2\times K_n}$, and its mode-3 unfolding matrix be $\mathbf{Y}_{(3)} \in \mathbb{R}^{K_n\times I_1 I_2}$. Then, $\mathfrak{L}$ is defined as

$$\mathfrak{L}(\mathcal{Y}) = \mathcal{Y} \times_3 \mathbf{U}_{\mathbf{Y}_{(3)}} \tag{12}$$

where $\mathbf{U}_{\mathbf{Y}_{(3)}} \in \mathbb{R}^{K_n\times K_n}$ is the left orthogonal matrix from the SVD of $\mathbf{Y}_{(3)}$ in real-time, satisfying $\mathbf{U}_{\mathbf{Y}_{(3)}}^T \mathbf{U}_{\mathbf{Y}_{(3)}} = \mathbf{I}$, i.e.,

$$\left[\mathbf{U}_{\mathbf{Y}_{(3)}}^t, , \right] = \mathrm{SVD}\left(\mathbf{Y}_{(3)}^t\right) \tag{13}$$

where t is the iteration number. The transformation basis is updated in real-time during optimization. Furthermore, the inverse HPT $\mathfrak{F}^{-1}$ is mathematically well-defined, provided that each explicit-layer matrix $\mathbf{S}_i$ is invertible. Since $\mathfrak{F} = \mathfrak{L} \circ \mathfrak{S}$, the inverse follows the standard composition rule $\mathfrak{F}^{-1} = \mathfrak{S}^{-1} \circ \mathfrak{L}^{-1}$. Noting that $\mathbf{U}_{\mathbf{Y}_{(3)}}$ is orthogonal by construction (cf. Remark 1), the original tensor can be recovered from the transformed tensor $\widetilde{\mathcal{X}} = \mathfrak{F}(\mathcal{X})$ as

$$\mathcal{X} = \mathfrak{F}^{-1}\left(\widetilde{\mathcal{X}}\right) = \widetilde{\mathcal{X}} \times_3 \mathbf{U}_{\mathbf{Y}_{(3)}}^T \times_3 \mathbf{S}_n^{-1} \times_3 \cdots \times_3 \mathbf{S}_1^{-1}. \tag{14}$$

Throughout this article, all definitions and derivations are presented in $\mathbb{R}$ for notational clarity. When a complex-valued transform (e.g., DFT) is employed in the explicit layer, $\mathbb{R}$ generalizes to $\mathbb{C}$, and the transpose $(\cdot)^T$ is understood as the conjugate transpose $(\cdot)^H$; all stated orthogonality properties and algebraic identities carry over without modification.

It is worth noting that the orthogonal basis $\mathbf{U}_{\mathbf{Y}_{(3)}}$ in the implicit layer $\mathfrak{L}$ is updated at every optimization iteration, synchronized with the intermediate transformed tensor.

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Specifically, at iteration t, the update follows a deterministic per-iteration mechanism

$$\mathbf{U}^t \leftarrow \text{left singular vectors of } \mathbf{Y}^t_{(3)}.$$

This indicates that no residual-based threshold or periodic schedule is employed. Instead, this synchronized continuous update inherently couples the transform adaptation to the optimization trajectory, ensuring that the adaptive basis continuously aligns with the evolving residual subspace and nonstationary structures.

Remark 1 (Mathematical Decoupling and Orthogonality of HPT): Mathematically, the decoupling mechanism of HPT fundamentally differs from a simple serial concatenation of transforms. If HPT were a mere algebraic chaining, the explicit bases and the learnable weights would collapse into a single unconstrained transformation matrix, suffering from the same ill-posed optimization as purely data-driven methods. Instead, HPT achieves rigorous mathematical decoupling through a *precondition-then-align* paradigm.

The explicit layer $\mathfrak{S}$ acts as a robust mathematical preconditioner. The standard DFT and DCT matrices are (scaled) orthogonal by construction: specifically, the unnormalized DFT matrix $\mathbf{F} \in \mathbb{C}^{N\times N}$ satisfies $\mathbf{F}^T\mathbf{F} = N\mathbf{I}$ (with the convention that $(\cdot)^T$ denotes $(\cdot)^H$ for complex matrices, as stated above), and the orthonormal DCT-II matrix $\mathbf{C}$ satisfies $\mathbf{C}^T\mathbf{C} = \mathbf{I}$. These properties directly satisfy the scaling orthogonality assumption in (20) with $\gamma_i = N$ for DFT and $\gamma_i = 1$ for DCT. By applying these fixed orthogonal transforms, $\mathfrak{S}$ dictates the global frequency decay, concentrating the dominant spatiospectral energy and forcing the intermediate tensor $\mathcal{Y} = \mathfrak{S}(\mathcal{X})$ into a spectrally concentrated, well-conditioned latent space.

Subsequently, the implicit layer $\mathfrak{L}$ does not learn a static, unconstrained weight matrix; rather, it computes an iteration-dependent orthogonal basis $\mathbf{U}$ via real-time SVD explicitly on the unfolded representation of this preconditioned tensor. Because $\mathbf{U}$ rigorously comprises the orthogonal left singular vectors, as established in the classic t-SVD framework [47], it inherently satisfies the strict orthogonal constraint (i.e., $\mathbf{U}^T\mathbf{U} = \mathbf{I}$). Thus, $\mathbf{U}$ functions purely as an adaptive subspace rotation rather than a free parameter matrix. This intrinsic orthogonality acts as a rigorous structural regularizer that strictly bounds the search space, effectively avoiding the overfitting and optimization instability typically encountered in purely data-driven learning mechanisms. Furthermore, this intrinsic orthogonality also satisfies the scaling orthogonality assumption with $\gamma = 1$, ensuring convergence stability.

As a direct consequence, the general inverse HPT in (14) admits a computationally efficient closed form that avoids explicit matrix inversions: each $\mathbf{S}_i^{-1}$ reduces to $\gamma_i^{-1}\mathbf{S}_i^T$ (with $\gamma_i = N$ for DFT and $\gamma_i = 1$ for DCT), while the implicit-layer inverse is simply $\mathbf{U}^T$.

Ultimately, this guarantees nonoverlapping mathematical roles: $\mathfrak{S}$ anchors the global energy distribution by dictating the frequency-domain decay profile, while $\mathfrak{L}$ aligns the representation with the principal directions of the residual covariance that fixed transforms cannot efficiently capture.

Definition 6 (Multidomain): Given an HPT $\mathfrak{F}(\cdot)$ as defined in Definition 5, the multidomain is denoted as the collection of transformed tensors under $\mathfrak{F}(\cdot)$, i.e.,

$$\text{Multi-domain} = \left\{\mathfrak{F}(\mathcal{X}) \mid \mathcal{X} \in \mathbb{R}^{I_1\times I_2\times I_3}\right\}. \tag{15}$$

Based on the concepts of the HPT and multidomain introduced above, we further define $\mathfrak{F}$-oriented tensor operations and related properties to create the foundation for subsequent model construction.

Definition 7 (HPT-Based t-Product): For arbitrary tensors $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times K_0}$ and $\mathcal{Y} \in \mathbb{R}^{I_2\times I_3\times K_0}$, the HPT $\mathfrak{F}$-oriented t-product $\mathcal{X} *_{\mathfrak{F}} \mathcal{Y} \in \mathbb{R}^{I_1\times I_3\times K_0}$ is defined by performing frontal slicewise matrix multiplication in the HPT domain, followed by the inverse HPT $\mathfrak{F}^{-1}$

$$\mathcal{X} *_{\mathfrak{F}} \mathcal{Y} = \mathfrak{F}^{-1}\left(\mathfrak{F}(\mathcal{X}) \odot \mathfrak{F}(\mathcal{Y})\right) \tag{16}$$

where $\odot$ denotes the frontal slicewise matrix multiplication. Consequently, this satisfies the mapping property $\mathfrak{F}(\mathcal{X} *_{\mathfrak{F}} \mathcal{Y}) = \mathfrak{F}(\mathcal{X}) \odot \mathfrak{F}(\mathcal{Y})$.

Definition 8 (HPT-Based Conjugate Transpose): The transpose of a tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times I_3}$ under $\mathfrak{F}$, denoted $\mathfrak{F}(\mathcal{X}^T)$, satisfies as $\mathfrak{F}(\mathcal{X}^T)^{(i)} = (\mathfrak{F}(\mathcal{X})^{(i)})^T, i = 1, 2, \ldots, K_n$, ensuring consistency of the transpose operation within the multidomain.

Definition 9 (HPT-Based Identity Tensor): For a given HPT $\mathfrak{F}$, a tensor $\mathcal{I} \in \mathbb{R}^{r\times r\times I_3}$ is termed an identity tensor under $\mathfrak{F}$ if all its frontal slices, after transformation via hierarchicaltransform, are identity matrices, i.e., $\mathfrak{F}(\mathcal{I})^{(i)} = \mathbf{I}_r$, where $\mathbf{I}_r$ is the $r \times r$ identity matrix.

Definition 10 (HPT-Based Orthogonal Tensor): A tensor $\mathcal{U} \in \mathbb{R}^{I\times I\times I_3}$ is orthogonal under $\mathfrak{F}$ if it satisfies $\mathcal{U}^T *_{\mathfrak{F}} \mathcal{U} = \mathcal{I} = \mathcal{U} *_{\mathfrak{F}} \mathcal{U}^T = \mathcal{I}$, with $\mathcal{I}$ being the identity tensor under $\mathfrak{F}$.

Definition 11 (HPT-Based t-SVD): For any tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times I_3}$, its $\mathfrak{F}$-based t-SVD is given by

$$\mathcal{X} = \mathcal{U} *_{\mathfrak{F}} \mathcal{S} *_{\mathfrak{F}} \mathcal{V}^T \tag{17}$$

where $\mathcal{U} \in \mathbb{R}^{I_1\times I_1\times I_3}$ and $\mathcal{V} \in \mathbb{R}^{I_2\times I_2\times I_3}$ are orthogonal tensors under $\mathfrak{F}$, and $\mathcal{S} \in \mathbb{R}^{I_1\times I_2\times I_3}$ is an f-diagonal tensor (each frontal slice $\mathcal{S}^{(i)}$ is a diagonal matrix).

Definition 12 ($\mathfrak{F}$-Oriented Tensor Rank): Given a tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times I_3}$ and its $\mathfrak{F}$-based t-SVD, its $\mathfrak{F}$-oriented rank is defined as the count of nonzero singular tubes in $\mathcal{S}$, i.e.,

$$\text{rank}_{\mathfrak{F}}(\mathcal{X}) := \#\{i : \mathcal{S}(i, i, :) \neq 0\} \tag{18}$$

where $\mathcal{S}(i, i, :)$ denotes the ith singular tube of $\mathcal{S}$.

The tensor $\mathfrak{F}$-oriented rank effectively captures the low-rank characteristics of tensors within the $\mathfrak{F}$-based transform domain. However, like most rank metrics, the $\mathfrak{F}$-rank is nonconvex and discrete, rendering direct minimization in optimization frameworks computationally challenging. To resolve this, we introduce a convex relaxation of the $\mathfrak{F}$-oriented rank—the learnable $\mathfrak{F}$-oriented TNN (LTNN), derived using the $\mathfrak{F}$-oriented tensor mode-k norm as follows.

Definition 13 ($\mathfrak{F}$-Oriented Tensor Mode-k Norm): Given that the HPT$\mathfrak{F}$ can be applied separately along the three dimensions, we denote the transform along the kth mode as $\mathfrak{F}_k$. Then, for a tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times K_0}$ and a specified mode k, $\mathfrak{F}$-oriented tensor spectral norm along mode k is defined as

$$\|\mathcal{X}\|_{\sigma,\mathfrak{F}_k} := \|\overline{\mathbf{X}}_{\mathfrak{F}_k}\|_\sigma \tag{19}$$

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where $\overline{\mathbf{X}}_{\mathfrak{F}_k} = \text{bdiag}(\widetilde{\mathcal{X}}_{\mathfrak{F}_k})$, with $\widetilde{\mathcal{X}}_{\mathfrak{F}_k} = \mathfrak{F}_k(\mathcal{X}) \in \mathbb{R}^{I_1 \times I_2 \times K_n}$ denoting the block diagonal matrix constructed from the mode-k slices of $\mathfrak{F}_k(\mathcal{X})$, and $\|\cdot\|_\sigma$ represents the matrix spectral norm, i.e., the largest singular value of the matrix. When focusing on the third mode—the spectral dimension in HSIs as the primary case—this simplifies to $\|\mathcal{X}\|_{\sigma,\mathfrak{F}_3}$ with corresponding adjustments to the block diagonal matrix dimensions based on the transformed tensor structure.

For tensors $\mathcal{X}, \mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times K_0}$ and the $\mathfrak{F}$-oriented spectral norm $\|\cdot\|_{\sigma,\mathfrak{F}_k}$ along mode k (see Definition 13), the dual norm leads to the LTNN, as derived in the following. HPT$\mathfrak{F}_k = \mathfrak{L}_k \circ \mathfrak{S}_k = \mathfrak{L}_k \circ \mathfrak{S}_{n,k} \circ \mathfrak{S}_{n-1,k} \circ \cdots \circ \mathfrak{S}_{i,k} \circ \mathfrak{S}_{1,k}$, where the ith linear transform along the kth mode $\mathfrak{S}_{i,k}$ and the learned transform $\mathfrak{L}$, respectively, correspond to the transform matrices $\mathbf{S}_{i,k}$ and $\mathbf{U}_k$. For $k = 1, 2, 3$, it is known that $\mathbf{U}_k^T\mathbf{U}_k = \mathbf{U}_k\mathbf{U}_k^T = \mathbf{I}$, and we assume that

$$\mathbf{S}_{i,k}^T\mathbf{S}_{i,k} = \mathbf{S}_{i,k}\mathbf{S}_{i,k}^T = \gamma_i\mathbf{I}, \quad i = 1, 2, \ldots, n_k \tag{20}$$

where n_k denotes the total number of explicit linear transforms cascaded along mode k and $\gamma_i > 0, i = 1, 2, \ldots, n_k$, are the specific constants. Consequently, we derive that

$$\langle \mathcal{X}, \mathcal{Y} \rangle = \frac{1}{\prod_{i=1}^{n_k} \gamma_i} \left\langle \widetilde{\mathcal{X}}_{\mathfrak{F}_k}, \widetilde{\mathcal{Y}}_{\mathfrak{F}_k} \right\rangle = \frac{1}{\prod_{i=1}^{n_k} \gamma_i} \left\langle \overline{\mathbf{X}}_{\mathfrak{F}_k}, \overline{\mathbf{Y}}_{\mathfrak{F}_k} \right\rangle. \tag{21}$$

From this, the LTNN of $\mathcal{X}$ along mode-k is

$$\begin{aligned} \|\mathcal{X}\|_{\star,\mathfrak{F}_k} &:= \sup_{\|\mathcal{Y}\|_{\sigma,\mathfrak{F}_k} \le 1} \langle \mathcal{X}, \mathcal{Y} \rangle = \sup_{\|\overline{\mathbf{Y}}_{\mathfrak{F}_k}\|_\sigma \le 1} \frac{1}{\prod_{i=1}^{n} \gamma_i} \left\langle \overline{\mathbf{X}}_{\mathfrak{F}_k}, \overline{\mathbf{Y}}_{\mathfrak{F}_k} \right\rangle \\ &\le \sup_{\|\mathbf{Z}\|_\sigma \le 1} \frac{1}{\prod_{i=1}^{n} \gamma_i} \left\langle \overline{\mathbf{X}}_{\mathfrak{F}_k}, \mathbf{Z} \right\rangle \\ &= \frac{1}{\prod_{i=1}^{n} \gamma_i} \left\| \overline{\mathbf{X}}_{\mathfrak{F}_k} \right\|_* \end{aligned} \tag{22}$$

where $\mathbf{Z} \in \mathbb{R}^{I_1 K_n \times I_2 K_n}$ is an arbitrary matrix. Meanwhile, $\mathcal{X}$ can be decomposed as $\mathcal{X} = \mathcal{U} *_{\mathfrak{F}} \mathcal{S} *_{\mathfrak{F}} \mathcal{V}^T$, so we also have

$$\begin{aligned} \|\mathcal{X}\|_{\star,\mathfrak{F}_k} &:= \sup_{\|\mathcal{Y}\|_\sigma \le 1} \langle \mathcal{X}, \mathcal{Y} \rangle \ge \left\langle \mathcal{U} *_{\mathfrak{F}} \mathcal{S} *_{\mathfrak{F}} \mathcal{V}^T, \mathcal{U} *_{\mathfrak{F}} \mathcal{V}^T \right\rangle \\ &= \left\langle \mathcal{U}^T *_{\mathfrak{F}_k} \mathcal{U} *_{\mathfrak{F}_k} \mathcal{S}, \mathcal{V}^T *_{\mathfrak{F}_k} \mathcal{V} \right\rangle \\ &= \langle \mathcal{S}, \mathcal{I} \rangle = \frac{1}{\prod_{i=1}^{n} \gamma_i} \left\langle \overline{\mathbf{S}}, \overline{\mathbf{I}} \right\rangle = \frac{1}{\prod_{i=1}^{n} \gamma_i} \text{Tr}\left(\overline{\mathbf{S}}\right) \\ &= \frac{1}{\prod_{i=1}^{n} \gamma_i} \left\| \overline{\mathbf{X}}_{\mathfrak{F}_k} \right\|_*. \end{aligned} \tag{23}$$

Then, the definition of LTNN can be summarized as follows.

Definition 14 (LTNN): For any HPT $\mathfrak{F}$, the LTNN of $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times K_0}$ is formulated as follows:

$$\|\mathcal{X}\|_{\star,\mathfrak{F}_k} = \frac{1}{\prod_{i=1}^{n} \gamma_i} \left\langle \overline{\mathbf{S}}, \overline{\mathbf{I}} \right\rangle = \frac{1}{\prod_{i=1}^{n} \gamma_i} \left\| \overline{\mathbf{X}}_{\mathfrak{F}_k} \right\|_*. \tag{24}$$

A single-mode TNN regularizer captures correlations along only one tensor dimension. To fully exploit the low-rank structures across all three modes (spatial, temporal, and spectral), we extend LTNN to a full-scale framework.

Definition 15 (Full-Scale LTNN): Given an arbitrary third-order tensor $\mathcal{X} \in \mathbb{R}^{I_0 \times J_0 \times K_0}$, by applying the full-scale HPT $\mathfrak{F}$ across all three tensor modes, we obtain $\widetilde{\mathcal{X}} = \mathfrak{F}(\mathcal{X}) \in \mathbb{R}^{I_n \times J_n \times K_n}$. Then, the full-scale LTNN$\|\mathcal{X}\|_{\star,\mathfrak{F}}$ is defined as

$$\begin{aligned} \|\mathcal{X}\|_{\star,\mathfrak{F}} &= \lambda_1\|\mathcal{X}\|_{\star,\mathfrak{F}_1} + \lambda_2\|\mathcal{X}\|_{\star,\mathfrak{F}_2} + \lambda_3\|\mathcal{X}\|_{\star,\mathfrak{F}_3} \\ &= \frac{1}{\prod_{i=1}^{n_k} \gamma_i} \left(\lambda_1\|\overline{\mathbf{X}}_{\mathfrak{F}_1}\|_* + \lambda_2\|\overline{\mathbf{X}}_{\mathfrak{F}_2}\|_* + \lambda_3\|\overline{\mathbf{X}}_{\mathfrak{F}_3}\|_*\right) \end{aligned} \tag{25}$$

where $\gamma_i > 0, i = 1, 2, \ldots, n_k$, are the constants and $\lambda_{1,2,3}$ are three nonnegative tradeoff parameters used to balance the importance of low-rank priors across different modes. By integrating low-rank priors across all three modes, this extension enables more comprehensive feature regularization and enhances the model's representational capacity.

The full-scale LTNN defined above employs the nuclear norm $\|\cdot\|_*$, i.e., the sum of singular values $\sum_i \sigma_i$, as the convex surrogate of the rank function. While this is the tightest convex relaxation of the matrix rank, it penalizes all singular values with a uniform unit gradient ($\partial/\partial\sigma_i \equiv 1$), regardless of their magnitude. In HSI patch groups, the leading singular values encode dominant spatial structures and spectral principal components, whereas the trailing ones correspond to noise and residual variations. A uniform penalty, therefore, overshrinks these informative components, resulting in loss of edges, textures, and fine spectral signatures.

To overcome this limitation, we replace the nuclear norm with the log-sum function $\text{LS}(\mathbf{X}) = \sum_i \log(\sigma_i(\mathbf{X}) + \varepsilon)$ [48], [49]. Its gradient $1/(\sigma_i + \varepsilon)$ is a decreasing function of σ_i: small singular values receive strong penalization that promotes sparsity, while large singular values are penalized far less, thereby preserving essential features. This concave, nonconvex surrogate provides a tighter approximation to the rank function than the nuclear norm [48], and its scalar proximal subproblem admits a closed-form solution as established by Gong et al. [49] (see Theorem 1). Similar log-sum penalties have been successfully adopted in recent HSI-SR methods, including low tensor multirank regularization (LTMR) [18], generalized TNN (GTNN) [17], and log-sum-based full -scale CTNN (LFCTNN) [22], where ablation studies consistently confirm significant performance gains over their nuclear-norm counterparts. Based on this, we define the LLTNN as follows.

Definition 16 (Log-Sum-Based Full-Scale LTNN, LLTNN): Given any tensor $\mathcal{X} \in \mathbb{R}^{I_0 \times J_0 \times K_0}$, the LLTNN$\|\mathcal{X}\|_{\star,L\mathfrak{F}}$ is defined as follows:

$$\begin{aligned} \|\mathcal{X}\|_{\star,L\mathfrak{F}} &= \lambda_1\|\mathcal{X}\|_{\star,L\mathfrak{F}_1} + \lambda_2\|\mathcal{X}\|_{\star,L\mathfrak{F}_2} + \lambda_3\|\mathcal{X}\|_{\star,L\mathfrak{F}_3} \\ &= \frac{\lambda_1}{\prod_{i=1}^{n} \gamma_i} \sum_{i=1}^{I_n} LS\left(\widetilde{\mathcal{X}}(i,:,:)\right) \\ &\quad + \frac{\lambda_2}{\prod_{i=1}^{n} \gamma_i} \sum_{i=1}^{J_n} LS\left(\widetilde{\mathcal{X}}(:,i,:)\right) \\ &\quad + \frac{\lambda_3}{\prod_{i=1}^{n} \gamma_i} \sum_{i=1}^{K_n} LS\left(\widetilde{\mathcal{X}}(:,:,i)\right) \\ &= \frac{\lambda_1}{\prod_{i=1}^{n} \gamma_i} \sum_{I=1}^{I_n} \sum_{j} \log\left(\sigma_j\left(\widetilde{\mathcal{X}}(i,:,:)\right) + \epsilon\right) \\ &\quad + \frac{\lambda_2}{\prod_{i=1}^{n} \gamma_i} \sum_{I=1}^{J_n} \sum_{j} \log\left(\sigma_j\left(\widetilde{\mathcal{X}}(:,i,:)\right) + \epsilon\right) \\ &\quad + \frac{\lambda_3}{\prod_{i=1}^{n} \gamma_i} \sum_{i=1}^{K_n} \sum_{j} \log\left(\sigma_j\left(\widetilde{\mathcal{X}}(:,:,i)\right) + \epsilon\right) \end{aligned} \tag{26}$$

where $LS(\mathbf{X}) = \sum_j \log(\sigma_j(\mathbf{X}) + \epsilon)$ and ϵ is a small positive constant.

Subsequently, we present the solution of LLTNN.

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Theorem 1 (LTSVT [22]): For a given $\mathcal{Z} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ and parameters $\eta > 0$ and $0 < \varepsilon < \min((\eta)^{1/2}, \eta/\sigma_{\max})$ where $\sigma_{\max}$ denotes the largest singular value of any slice of $\mathcal{W}$, the solution to the regularization

$$\arg\min_{\mathcal{Z}} \frac{1}{2}\|\mathcal{Z}-\mathcal{W}\|_F^2 + \eta \sum_{i=1}^{I_1} \mathrm{LS}\left(\mathcal{Z}(i,:,:)\right) \tag{27}$$

$$\arg\min_{\mathcal{Z}} \frac{1}{2}\|\mathcal{Z}-\mathcal{W}\|_F^2 + \eta \sum_{i=1}^{I_2} \mathrm{LS}\left(\mathcal{Z}(:,i,:)\right) \tag{28}$$

$$\arg\min_{\mathcal{Z}} \frac{1}{2}\|\mathcal{Z}-\mathcal{W}\|_F^2 + \eta \sum_{i=1}^{I_3} \mathrm{LS}\left(\mathcal{Z}(:,:,i)\right) \tag{29}$$

is given by the following LTSVT operator:

$$LTSVT_{\eta,\epsilon}(\mathcal{Z}) = \mathcal{U} * \mathcal{S} * \mathcal{V}^T \tag{30}$$

where $\mathcal{U}$ and $\mathcal{V}$ are obtained by the SVD on the tensor $\mathcal{W}$, i.e., $\mathcal{W} = \mathcal{U} * \Sigma * \mathcal{V}^T$, and $\mathcal{S}$ can be calculated by

$$\mathcal{S}_{\eta,\varepsilon}(i,:,:) = \begin{cases} 0, & c_2 \le 0 \\ \mathrm{sign}\left(\Sigma(i,:,:)\right)\left(\frac{c_1+\sqrt{c_2}}{2}\right), & c_2 > 0 \end{cases} \tag{31}$$

$$\mathcal{S}_{\eta,\varepsilon}(:,i,:) = \begin{cases} 0, & c_2 \le 0 \\ \mathrm{sign}\left(\Sigma(:,i,:)\right)\left(\frac{c_1+\sqrt{c_2}}{2}\right), & c_2 > 0 \end{cases} \tag{32}$$

$$\mathcal{S}_{\eta,\varepsilon}(:,:,i) = \begin{cases} 0, & c_2 \le 0 \\ \mathrm{sign}\left(\Sigma(:,:,i)\right)\left(\frac{c_1+\sqrt{c_2}}{2}\right), & c_2 > 0. \end{cases} \tag{33}$$

In formula (32), $\Sigma(i,:,:)$ is the ith horizontal slice, $c_1 = |\mathcal{S}(i,:,:)| - \epsilon$, and $c_2 = (\mathcal{S}(i,:,:) + \epsilon)^2 - 4\eta$.

The proof is provided in the supplementary material.

B. Nonlocal LLTNN Regularization for Coefficients and Final Model

In Section III-C, we established the HSI-SR observation model (9). However, directly solving for $\mathcal{A}$ from these equations is a severely ill-posed problem. To stabilize the solution, we must incorporate priors that reflect the intrinsic structure of HSIs.

We exploit the spatial nonlocal self-similarity (NSS) prior. HSIs exhibit significant spatial redundancy, meaning that patches at different locations often share similar structures. Instead of regularizing the entire tensor $\mathcal{A}$, we apply constraints on groups of similar patches, which has proven to be a more powerful and computationally tractable approach [22].

Our procedure is given as follows.

1) *Clustering:* We leverage the HR-MSI $\mathcal{Z}$, which provides reliable spatial structures, to perform clustering. We partition $\mathcal{Z}$ into N_p overlapping spatial patches and group them into C clusters using K-means.
2) *Patch Group Construction:* Using the spatial indices from clustering $\mathcal{Z}$, we extract the corresponding patches from the coefficient tensor $\mathcal{A}$. All N_j patches belonging to the jth cluster are collected and stacked to form a 3-D tensor group $\mathcal{A}^j \in \mathbb{R}^{q \times L \times N_j}$ (where q is the spatial patch size, L is the coefficient dimension, and N_j is the number of patches in the jth group).

Since each tensor group $\mathcal{A}^j$ consists of mutually similar patches, it is expected to exhibit a strong low-rank structure across all three modes (spatial, coefficient, and nonlocal).

We, therefore, propose to regularize each patch group $\mathcal{A}^j$ using our *LLTNN* regularizer. This leads to our final *Nonlocal LLTNN* model for HSI-SR

$$\min_{\mathcal{A}} \left\|\mathbf{Y}_{(3)} - \mathbf{D}\mathbf{A}_{(3)}\mathbf{B}\mathbf{S}\right\|_F^2 + \left\|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{D}\mathbf{A}_{(3)}\right\|_F^2 + \sum_{j=1}^{C} \left\|\mathcal{A}^j\right\|_{\star,L\mathfrak{F}} \tag{34}$$

where the regularizer $\|\mathcal{A}^j\|_{\star,L\mathfrak{F}}$ is the full-scale LLTNN defined in (16)

$$\left\|\mathcal{A}^j\right\|_{\star,L\mathfrak{F}} = \lambda_1 \left\|\mathcal{A}^j\right\|_{\star,L\mathfrak{F}_1} + \lambda_2 \left\|\mathcal{A}^j\right\|_{\star,L\mathfrak{F}_2} + \lambda_3 \left\|\mathcal{A}^j\right\|_{\star,L\mathfrak{F}_3}.$$

This model (34) effectively combines the spectral subspace prior with a powerful nonlocal regularization on the coefficients, which is constrained by our novel LLTNN framework. To better illustrate our model, a flowchart is presented in Fig. 2.

V. Optimization via ADMM

We solve the proposed LLTNN model (34) using the ADMM framework. The objective function is complex, nonconvex, and nonsmooth due to the nonlocal patch-based LLTNN regularizer.

We first introduce auxiliary variables $\mathcal{U}_1, \mathcal{U}_2, \mathcal{U}_3 \in \mathbb{R}^{H \times W \times L}$, forcing them to be equal to the coefficient tensor $\mathcal{A}$. This allows us to decouple the data-fidelity terms from the regularization terms. The optimization problem in (34) is rewritten as

$$\begin{aligned} \min_{\mathbf{A},\{\mathcal{U}_k\}} & \left\|\mathbf{Y}_{(3)} - \mathbf{D}\mathbf{A}_{(3)}\mathbf{B}\mathbf{S}\right\|_F^2 + \left\|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{D}\mathbf{A}_{(3)}\right\|_F^2 \\ & + \sum_{j=1}^{C} \left\|\mathcal{U}_k^j\right\|_{\star,L\mathfrak{F}_k} \\ \text{s.t.} \quad & \mathcal{A} = \mathcal{U}_k, \quad k = 1,2,3 \end{aligned} \tag{35}$$

where

$$\left\|\mathcal{A}^j\right\|_{\star,L\mathfrak{F}} = \lambda_1 \left\|\mathcal{A}^j\right\|_{\star,L\mathfrak{F}_1} + \lambda_2 \left\|\mathcal{A}^j\right\|_{\star,L\mathfrak{F}_2} + \lambda_3 \left\|\mathcal{A}^j\right\|_{\star,L\mathfrak{F}_3}$$

is the nonlocal LLTNN regularizer applied to the jth patch group $\{\mathcal{U}_k^j\}$ extracted from $\mathcal{U}_k$.

The augmented Lagrangian function $\mathcal{L}_\rho$ for this problem is

$$\begin{aligned} &\mathcal{L}_\rho(\mathcal{A},\{\mathcal{U}_k\},\{Q_k\}) \\ &= \left\|\mathbf{Y}_{(3)} - \mathbf{D}\mathbf{A}_{(3)}\mathbf{B}\mathbf{S}\right\|_F^2 + \left\|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{D}\mathbf{A}_{(3)}\right\|_F^2 \\ &\quad + \rho \sum_{k=1}^{3} \left\|\mathcal{U}_k - \mathcal{A} + \frac{\mathcal{Q}_k}{2\rho}\right\|_F^2 + \sum_{k=1}^{3}\sum_{j=1}^{C} \lambda_k \left\|\mathcal{U}_k^j\right\|_{\star,L\mathfrak{F}_k} \end{aligned} \tag{36}$$

where Q_k are the Lagrangian multipliers (dual variables) and $\rho > 0$ are penalty parameters. The ADMM algorithm iteratively solves the following subproblems.

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A. Update $\mathcal{A}$—Coefficient Subproblem

At iteration $t+1$, we fix $\mathcal{U}_k = \mathcal{U}_k^t$ and $Q_k = Q_k^t$ to update $\mathcal{A}$

$$\mathcal{A}^{t+1} = \arg\min_{\mathcal{A}} \left\|\mathbf{Y}_{(3)} - \mathbf{D}\mathbf{A}_{(3)}\mathbf{B}\mathbf{S}\right\|_F^2 + \left\|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{D}\mathbf{A}_{(3)}\right\|_F^2 + \rho\sum_{k=1}^{3}\left\|\mathcal{U}_k^t - \mathcal{A}^t + \frac{\mathcal{Q}_k^t}{2\rho}\right\|_F^2. \tag{37}$$

This is a large-scale least-squares problem. By taking the derivative with respect to $\mathbf{A}_{(3)}$ and setting it to zero, we obtain a standard Sylvester equation

$$\mathbf{H}_1\mathbf{A}_{(3)} + \mathbf{A}_{(3)}\mathbf{H}_2 = \mathbf{H}_3 \tag{38}$$

where

$$\mathbf{H}_1 = (\mathbf{R}\mathbf{D})^T(\mathbf{R}\mathbf{D}) + \rho\mathbf{I}_L \tag{39}$$

$$\mathbf{H}_2 = (\mathbf{B}\mathbf{S})(\mathbf{B}\mathbf{S})^T \tag{40}$$

$$\mathbf{H}_3 = (\mathbf{R}\mathbf{D})^T\mathbf{Z}_{(3)} + \mathbf{D}^T\mathbf{Y}_{(3)}(\mathbf{B}\mathbf{S})^T + \sum_{k=1}^{3}\left(\rho\mathbf{U}_{k(3)}^t + \mathbf{Q}_{k(3)}^t\right). \tag{41}$$

Algorithm 1 Update $\mathcal{A}$ via the Sylvester Equation

Require: $\mathbf{Y}_{(3)}, \mathbf{Z}_{(3)}, \mathbf{D}, \mathbf{R}, \mathbf{B}, \mathbf{S}$; current $\{\mathcal{U}_k^t, \mathcal{Q}_k^t, \}_{k=1}^3, \rho$
Ensure: $\mathcal{A}^{t+1}$

Construct: $\mathbf{H}_1 \leftarrow (\mathbf{R}\mathbf{D})^\top(\mathbf{R}\mathbf{D}) + \rho\,\mathbf{I}_L$,
$\mathbf{H}_2 \leftarrow (\mathbf{B}\mathbf{S})(\mathbf{B}\mathbf{S})^\top$,
$\mathbf{H}_3 \leftarrow (\mathbf{R}\mathbf{D})^\top\mathbf{Z}_{(3)} + \mathbf{D}^\top\mathbf{Y}_{(3)}(\mathbf{B}\mathbf{S})^\top + \sum_{k=1}^{3}(\rho\mathcal{U}_k^t - \mathcal{Q}_k^t)_{(3)}$.
Solve the Sylvester equation:
$\mathbf{H}_1\,\mathbf{A}_{(3)} + \mathbf{A}_{(3)}\,\mathbf{H}_2 = \mathbf{H}_3$, for $\mathbf{A}_{(3)}$ using a fast Sylvester solver.
Fold back: $\mathcal{A}^{t+1} \leftarrow \text{fold}_3(\mathbf{A}_{(3)})$.

This equation can be solved efficiently using the fast Sylvester equation solver described in [17]. The detailed procedure for updating the coefficient tensor is summarized in Algorithm 1.

B. Update $\mathcal{U}_k$—Regularization Subproblem

The updates for $\mathcal{U}_k$ are decoupled and can be solved in parallel for $k = 1, 2, 3$

$$\mathcal{U}_k^{t+1} = \arg\min_{\mathcal{U}_k} \lambda_k\sum_{j=1}^{C}\left\|\mathcal{U}_k^{t,j}\right\|_{\star,L\mathfrak{F}_k} + \rho\sum_{k=1}^{3}\left\|\mathcal{U}_k^t - \mathcal{A}^t + \frac{\mathcal{Q}_k^t}{2\rho}\right\|_F^2. \tag{42}$$

The above equation involves low-rank tensor regularization optimization, and we can solve it by dealing with each cluster one by one. Taking $\mathcal{U}_1$ and the jth cluster as an example, we have

$$\arg\min_{\mathcal{U}_1^{t+1,j}} \lambda_1\sum_{j=1}^{N_j} LS\left(\mathcal{U}_1^{t,j}(i,:,:)\right) + \frac{\rho}{N_j}\left\|\mathcal{U}_1^{j,t} - \mathcal{W}_1^{t,j}\right\|_F \tag{43}$$

where N_j is the number of patches in the jth class. Since each HPT $\mathfrak{F}_k$ consists of (scaled) orthogonal transforms (cf. Remark 1), the Frobenius norm satisfies $\|\mathcal{Z} - \mathcal{W}\|_F^2 = \left(\prod_i \gamma_i\right)^{-1}\|\widetilde{\mathcal{Z}} - \widetilde{\mathcal{W}}\|_F^2$. Absorbing the scaling constant into the penalty parameter, (43) reduces to the standard proximal form of the LLTNN regularizer (applied along mode-1). The solution is obtained by applying the *LTSVT* operator (defined in Theorem 1) to each patch group $\mathcal{W}_1^{t,j}$. The complete parallel updating process for the nonlocal regularization variables $\{\mathcal{U}_k\}_{k=1}^3$ is outlined in Algorithm 2.

Algorithm 2 Update $\{\mathcal{U}_k\}_{k=1}^3$ via LTSVT on Nonlocal Groups

Require: $\mathcal{A}^{t+1}$; ρ; $\{Q_k^t, \lambda_k\}_{k=1}^3$; grouping indices $\{\mathcal{I}_j\}_{j=1}^C$
Ensure: $\{\mathcal{U}_k^{t+1}\}_{k=1}^3$

for $k = 1$ to 3 **in parallel do**
  $\mathcal{W}_k^t \leftarrow \mathcal{A}^{t+1} + \mathcal{Q}_k^t/(2\rho)$
  for $j = 1$ to C **do**
    Extract patch group: $\mathcal{W}_k^{t,j}$ using indices $\mathcal{I}_j$
    $\eta \leftarrow \lambda_k/\rho$
    $\mathcal{U}_k^{t+1,j} \leftarrow \text{LTSVT}_{\eta,\epsilon}(\mathcal{W}_k^{t,j})$
  end for
  $\mathcal{U}_k^{t+1} \leftarrow \text{GroupAggregate}(\{\mathcal{U}_k^{t+1,j}\}_{j=1}^C; \{\mathcal{I}_j\})$
end for

Algorithm 3 ADMM for LLTNN-Based HSI-SR

Require: $\mathbf{Y}_{(3)}, \mathbf{Z}_{(3)}$; operators $\mathbf{D}, \mathbf{R}, \mathbf{B}, \mathbf{S}$; groups $\{\mathcal{I}_j\}_{j=1}^C$; ρ; $\{\lambda_k\}_{k=1}^3$; tolerance tol
Ensure: Coefficient tensor $\mathcal{A}$

Initialize $\mathcal{A}^0$ via least-squares: $\mathcal{A}^0 \leftarrow \arg\min_{\mathcal{A}}$ Eq. (9)
$\mathcal{U}_k^0 \leftarrow \mathcal{A}^0$, $\mathcal{Q}_k^0 \leftarrow \mathbf{0}$ for $k = 1, 2, 3$; $t \leftarrow 0$
while not converged **do**
  A-update: $\mathcal{A}^{t+1} \leftarrow$ Algorithm 1
  U-update: $\{\mathcal{U}_k^{t+1}\}_{k=1}^3 \leftarrow$ Algorithm 2
  Dual update: $\mathcal{Q}_k^{t+1} \leftarrow \mathcal{Q}_k^t + \rho(\mathcal{A}^{t+1} - \mathcal{U}_k^{t+1})$, $k = 1, 2, 3$
  Compute $r^{t+1} \leftarrow \sum_k \|\mathcal{A}^{t+1} - \mathcal{U}_k^{t+1}\|_F$
  if $r^{t+1} \le$ tol **then break**
  end if
  $t \leftarrow t + 1$
end while
return $\mathcal{A}^{t+1}$

C. Update Q_k—Multiplier Subproblem

Finally, the Lagrangian multipliers are updated

$$\mathcal{Q}_k^{t+1} = \mathcal{Q}_k^t + \rho_k\left(\mathcal{A}^{t+1} - \mathcal{U}_k^{t+1}\right), \quad k = 1, 2, 3. \tag{44}$$

We iterate steps 1–3 until convergence. The overall ADMM optimization framework for the proposed LLTNN-based HSI-SR model is systematically summarized in Algorithm 3.

D. Computational Complexity

We analyze the per-iteration cost of Algorithm 3. Let H and W denote the spatial dimensions, L the subspace dimension, C the number of nonlocal clusters, q the spatial patch size, and $N_m = \max_j N_j$ the maximum cluster size.

The $\mathcal{A}$-subproblem solves the Sylvester equation (38) via FFT-based diagonalization, costing $O(HW\max\{\log(HW), L^2\})$—identical to GTNN [17] and LFCTNN [22]. The $\mathcal{U}_k$-subproblem applies the HPT and LTSVT operator to each patch group $\mathcal{A}^j \in \mathbb{R}^{q\times L\times N_j}$

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TABLE I

PER-ITERATION COMPLEXITY OF THE COMPARED METHODS. †NOT PROVIDED IN THE ORIGINAL ARTICLE; DERIVED FROM ITS ALGORITHMIC STRUCTURE

Method	Dominant per-iteration cost
CSTF [50]	$O\big(N_{\mathrm{CG}}(n_w^2 W + n_h^2 H) + n_w^2 n_h n_s\big)$
LTMR [18]	$O(HW\max\{\log(HW), S^2\}) + O(CqS^2 N_m)$
LTTR [51]	$O(HW\max\{\log(HW), S^2\}) + O(Cq^2 L N_m)^{\dagger}$
CTRF [52]	$O(N_{\mathrm{CG}} R_1 R_2 HW) + O(R_2 S \min(R_2, S))^{\dagger}$
GTNN [17]	$O(HW\max\{\log(HW), L^2\}) + O(CqL^2 N_m)$
LFCTNN [22]	$O(HW\max\{\log(HW), L^2\}) + O(CqLN_m(N_m + L))$
LLTNN	$O(HW\max\{\log(HW), L^2\}) + O(CqLN_m(N_m + L))$

across three modes. Taking mode-1 as an example, the cost per cluster comprises: 1) the explicit-layer transforms (DFT+DCT), costing $O(qLN_j \log q)$; 2) the implicit-layer SVD of the mode-1 unfolding, costing $O(q^2 LN_j)$; and 3) frontal-slice SVDs for LTSVT, costing $O(qLN_j \min(q, L))$. The DFT/DCT term is strictly lower-order than the SVD terms and does not affect the asymptotic bound. Modes 2 and 3 follow analogously by permuting the roles of q, L, and N_j. The dual update costs $O(HWL)$ and is negligible.

Table I compares the per-iteration complexity of all methods. LLTNN shares the same asymptotic complexity class as LFCTNN-FL, confirming that the HPT's explicit layer introduces no additional Big-O overhead, while accuracy gains demonstrated in Section VI are achieved at no extra cost.

VI. EXPERIMENTAL RESULTS

To evaluate the effectiveness and superiority of the proposed method in HSI-SR, we compare LLTNN with six state-of-the-art baselines: low tensor train rank representation (LTTR) [51], coupled sparse tensor factorization (CSTF) [50], LTMR [18], coupled tensor ring factorization (CTRF) [52], LFCTNN [22], SDP [34], and the GTNN [17] approach. For a fair comparison, we tune the relevant hyperparameters of each baseline within the ranges recommended in their original articles or official implementations and adopt a common validation protocol to obtain the best performance. The experiments are implemented using MATLAB R2020a on a Windows 11 system, with an Intel Core i9-14900K CPU (3.20 GHz) and 64 GB of RAM.

A. Simulated Datasets and Implementation Details

We use four widely adopted hyperspectral datasets with diverse characteristics: Pavia University (PU), Washington DC Mall (WDC), Cuprite (CU), and the Columbia Computer Vision Laboratory (CAVE) dataset.

1) PU Dataset: This dataset features a university campus scene, captured by the reflective optics system imaging spectrometer (ROSIS) sensor over the University of Pavia, Italy. The original image consists of 610 × 340 pixels with 103 spectral bands, a spectral range of 0.43–0.86 μm, and a spatial resolution of 1.3 m. After removing bands affected by water vapor absorption, 93 bands were retained. For our experiments, a subimage of size 320 × 320 × 93 was cropped to serve as the reference HSI.

2) WDC Dataset: This dataset was acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor over the Washington DC Mall. It originally comprises 210 bands covering the visible and infrared spectrum from 0.4 to 2.4 μm. The available data, after removing noisy and water absorption bands, has dimensions of 1280 × 307 × 191. From this, we cropped a spatial subregion of 256 × 256 pixels and selected 93 bands, resulting in a 256 × 256 × 93 reference image for evaluation.

3) CU Dataset: Collected over the Cuprite mining district in Nevada, USA, this dataset originally contains 224 spectral bands from 370 to 2480 nm with a spatial dimension of 614 × 512 pixels. Following a similar preprocessing procedure, the number of spectral bands was reduced to 93, and a 320 × 320 spatial area was cropped, yielding a reference image of size 320 × 320 × 93.

4) CAVE Dataset: This dataset consists of 32 indoor scenes. Each image has a spatial resolution of 512 × 512 pixels and contains 31 spectral bands spanning the 400–700-nm range. Due to the smaller number of native spectral bands, we utilized the full 31 bands and the original 512 × 512 spatial dimensions in our experiments.

To generate the LR-HSI and HR-MSI required for the fusion task, we simulated the degradation process from the reference HSIs described above. The LR-HSIs were generated by first applying a Gaussian blur (with a kernel of 7 × 7 and a standard deviation of 2) to the reference HSI, followed by spatial downsampling. The spatial downsampling factor was set to 4 for the PU, WDC, and CU datasets, and 16 for the CAVE dataset. The corresponding LR-MSIs were generated by spectrally degrading the reference HSIs using the spectral response functions (SRFs) of the IKONOS sensor for the first three datasets and the Nikon D700 camera for the CAVE dataset. Finally, zero-mean Gaussian noise was added to both the simulated LR-HSI and LR-MSI. The SNRs were set to fall within the range of 30–35 dB for the PU, WDC, and CU datasets, and fixed at 20 dB for the CAVE dataset. For the fusion algorithm itself, no external initialization is required: the auxiliary variables $\{\mathcal{U}_k\}_{k=1}^3$ and the Lagrange multipliers $\{\mathcal{Q}_k\}_{k=1}^3$ are initialized to zero so that the first iteration of 37 reduces to a least-squares problem in $\mathcal{A}$ driven purely by the LR-HSI and HR-MSI data fidelity terms. Its closed-form Sylvester solution yields the initial coefficient tensor $\mathcal{A}^1$, and the corresponding initial fused image is recovered as $\mathbf{X}_{(3)} = \mathbf{D}\mathbf{A}_{(3)}^1$. All subsequent updates are then carried out within the proposed ADMM framework.

B. Evaluation Metrics

We adopt six quantitative metrics to evaluate the quality of super-resolved images from complementary perspectives: peak signal-to-noise ratio (PSNR) [53], structural similarity index (SSIM) [54], correlation coefficient (CC) [55], spectral angle mapper (SAM) [56], relative dimensionless global error in synthesis (ERGAS) [57], and degree of distortion (DD) [50]. PSNR is used to measure the reconstruction error between the super-resolved image and the ground truth; higher values indicate better fidelity. SSIM combines luminance, contrast, and structure to quantify structural similarity, whereas CC

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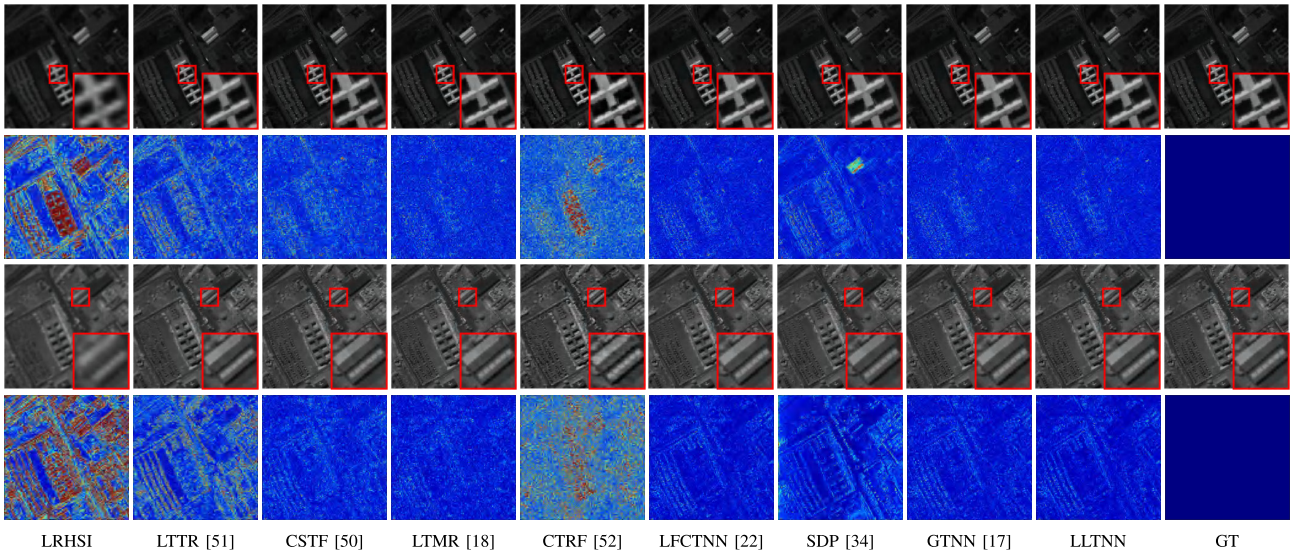


Fig. 3. Visual comparison of different methods on the PU dataset: the first two rows present the recovered HSIs and their corresponding error images at band 1; the last two rows show the recovered results and the error images at band 73.

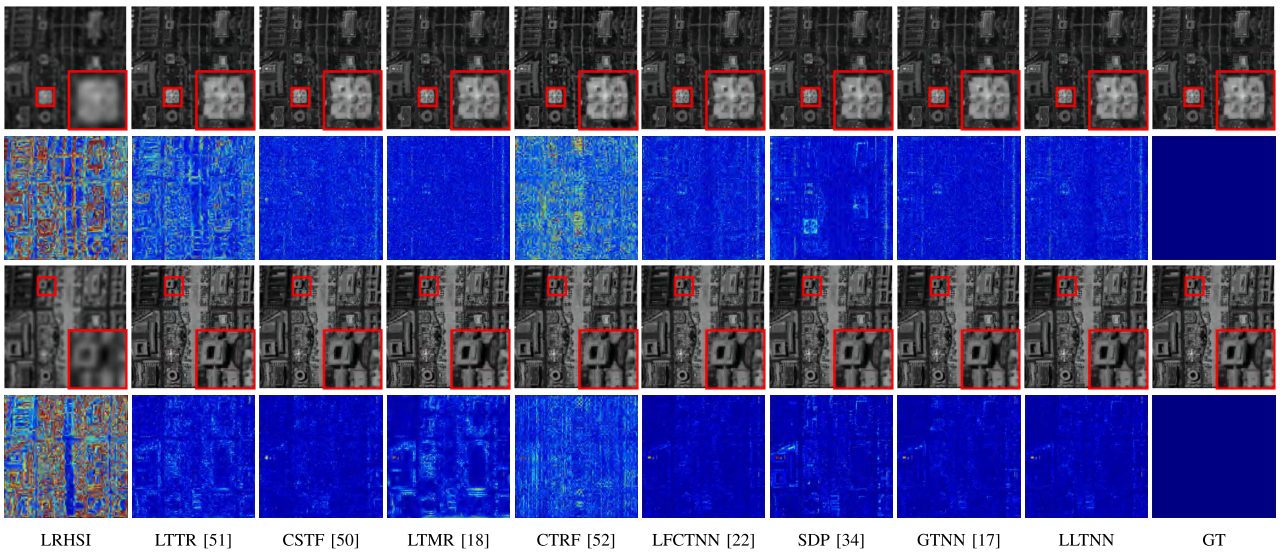


Fig. 4. Reconstruction HSIs' comparison on the WDC dataset: the upper two rows display the restored images and error images for the third spectral bands, and the lower two rows present the reconstructed results and the associated error maps at band 90.

reflects distributional consistency with the reference; for both metrics, values closer to 1 are preferred. SAM assesses spectral similarity in angular form, where smaller angles indicate less spectral distortion. ERGAS jointly reflects the consistency of spectral and spatial information, and DD measures discrepancies in image details. For these last two metrics (ERGAS and DD), lower values indicate better detail preservation. Taken together, these metrics provide a comprehensive and reliable assessment of HSI-SR performance.

C. Results on Simulated Data

In this section, we analyze the fusion results of seven models on the four datasets described above.

1) PU Dataset: Table II reports the quantitative results of all methods across all datasets; for clarity, the best and second-best scores are highlighted in red and blue, respectively. On PU, although LTMR, GTNN, and LFCTNN already deliver competitive reconstructions, the proposed LLTNN achieves the best performance across all reported metrics. In particular, LLTNN attains a PSNR of 43.7814 dB, representing a 0.48-dB improvement over the second-best LFCTNN. Fig. 3 presents visual comparisons on the first and 73rd bands of PU, including gray-scale reconstructions, error maps with respect to the ground truth, and two zoomed-in regions indicated by red boxes for direct inspection. From the enlarged views, all methods reduce artifacts to some extent; however, LTTR and CSTF exhibit noticeable texture loss and nonuniform gray-level distributions, and the remaining baselines still fail to recover rooftop details faithfully. By contrast, LLTNN preserves finer textures and produces more spatially uniform reconstructions.

2) WDC Dataset: From the metrics in Table II, LLTNN achieves higher reconstruction accuracy and shows a larger margin than LFCTNN and GTNN. To visualize the fusion outcome, gray-scale reconstructions on the third and 90th bands are presented in Fig. 4. From the annotated regions of

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TABLE II

QUANTITATIVE METRICS (PSNR, SSIM, CC, SAM, ERGAS, AND DD) OF ALL COMPARED APPROACHES. THE BEST VALUES AND THE SECOND BEST VALUES ARE, RESPECTIVELY, HIGHLIGHTED BY RED AND BLUE

Image	Method	PSNR↑	SSIM↑	CC↑	SAM↓	ERGAS↓	DD↓
PU	$\text{LTTR}_{\text{TNNLS19}}$	37.5980 ↓14.12%	0.9510 ↓3.69%	0.9862 ↓1.10%	4.2409 ↑122.44%	2.1797 ↑111.93%	2.6209 ↑117.66%
	$\text{CSTF}_{\text{TCYB22}}$	40.8219 ↓6.76%	0.9757 ↓1.18%	0.9949 ↓0.23%	2.6849 ↑40.83%	1.4385 ↑39.86%	1.7602 ↑46.18%
	$\text{LTMR}_{\text{TIP22}}$	42.6951 ↓2.48%	0.9828 ↓0.47%	0.9966 ↓0.06%	2.2473 ↑17.88%	1.1697 ↑13.73%	1.4085 ↑16.98%
	$\text{CTRF}_{\text{PR22}}$	36.1830 ↓17.36%	0.9363 ↓5.18%	0.9826 ↓1.46%	4.9983 ↑162.17%	2.6039 ↑153.17%	3.2867 ↑172.96%
	$\text{LFCTNN}_{\text{TRGS24}}$	43.3014 ↓1.10%	0.9858 ↓0.16%	0.9970 ↓0.02%	2.0337 ↑6.67%	1.0895 ↑5.93%	1.2851 ↑6.73%
	$\text{SDP}_{\text{TRGS24}}$	41.7399 ↓4.66%	0.9857 ↓0.17%	0.9959 ↓0.13%	2.2742 ↑19.29%	1.3055 ↑26.93%	1.5156 ↑26.15%
	$\text{GTNN}_{\text{TNNLS25}}$	43.1470 ↓1.45%	0.9850 ↓0.24%	0.9969 ↓0.03%	2.0924 ↑9.75%	1.1048 ↑7.42%	1.3189 ↑9.53%
	$\text{LLTNN}_{\text{Ours}}$	43.7814 ↓0.00%	0.9874 ↓0.00%	0.9972 ↓0.00%	1.9065 ↑0.00%	1.0285 ↑0.00%	1.2041 ↑0.00%
WDC	$\text{LTTR}_{\text{TNNLS19}}$	31.9492 ↓21.06%	0.9262 ↓6.24%	0.9765 ↓1.78%	8.0336 ↑164.87%	3.4587 ↑182.71%	5.4829 ↑185.75%
	$\text{CSTF}_{\text{TCYB22}}$	38.6877 ↓4.41%	0.9819 ↓0.60%	0.9961 ↓0.12%	3.9431 ↑30.01%	1.4856 ↑21.43%	2.4297 ↑26.63%
	$\text{LTMR}_{\text{TIP22}}$	36.8625 ↓8.92%	0.9707 ↓1.73%	0.9930 ↓0.43%	4.7450 ↑56.45%	1.9801 ↑61.85%	3.1370 ↑63.49%
	$\text{CTRF}_{\text{PR22}}$	30.2050 ↓25.37%	0.8972 ↓9.17%	0.9718 ↓2.56%	10.8474 ↑257.65%	4.3453 ↑255.18%	7.4618 ↑288.88%
	$\text{LFCTNN}_{\text{TRGS24}}$	39.9451 ↓1.30%	0.9871 ↓0.07%	0.9971 ↓0.02%	3.2847 ↑8.30%	1.2737 ↑4.11%	1.9964 ↑4.04%
	$\text{SDP}_{\text{TRGS24}}$	38.4466 ↓5.01%	0.9856 ↓0.22%	0.9963 ↓0.10%	3.0971 ↑2.11%	1.4911 ↑21.88%	2.3183 ↑20.82%
	$\text{GTNN}_{\text{TNNLS25}}$	39.8100 ↓1.64%	0.9864 ↓0.14%	0.9970 ↓0.03%	3.2456 ↑7.01%	1.2996 ↑6.23%	2.0888 ↑8.86%
	$\text{LLTNN}_{\text{Ours}}$	40.4722 ↓0.00%	0.9878 ↓0.00%	0.9973 ↓0.00%	3.0330 ↑0.00%	1.2234 ↑0.00%	1.9188 ↑0.00%
CU	$\text{LTTR}_{\text{TNNLS19}}$	34.0196 ↓25.73%	0.8542 ↓13.41%	0.9510 ↓4.55%	3.3608 ↑399.97%	1.8430 ↑276.81%	3.9255 ↑302.78%
	$\text{CSTF}_{\text{TCYB22}}$	40.1726 ↓12.30%	0.9514 ↓3.56%	0.9852 ↓1.11%	1.6300 ↑142.49%	1.0213 ↑108.81%	1.9723 ↑102.37%
	$\text{LTMR}_{\text{TIP22}}$	41.0375 ↓10.41%	0.9575 ↓2.94%	0.9878 ↓0.85%	1.4599 ↑117.18%	0.9076 ↑85.57%	1.7821 ↑82.85%
	$\text{CTRF}_{\text{PR22}}$	42.5129 ↓7.19%	0.9762 ↓1.04%	0.9932 ↓0.31%	0.8382 ↑24.70%	0.6645 ↑35.86%	1.4610 ↑49.91%
	$\text{LFCTNN}_{\text{TRGS24}}$	42.8084 ↓6.55%	0.9719 ↓1.48%	0.9922 ↓0.41%	1.1407 ↑69.70%	0.7179 ↑46.78%	1.4395 ↑47.70%
	$\text{SDP}_{\text{TRGS24}}$	42.5746 ↓7.06%	0.9845 ↓0.20%	0.9927 ↓0.36%	1.0725 ↑59.55%	0.6961 ↑42.32%	1.4414 ↑47.90%
	$\text{GTNN}_{\text{TNNLS25}}$	42.1961 ↓7.88%	0.9670 ↓1.98%	0.9908 ↓0.55%	1.2457 ↑85.32%	0.7811 ↑59.70%	1.5506 ↑59.10%
	$\text{LLTNN}_{\text{Ours}}$	45.8066 ↓0.00%	0.9865 ↓0.00%	0.9963 ↓0.00%	0.6722 ↑0.00%	0.4891 ↑0.00%	0.9746 ↑0.00%
CAVE	$\text{LTTR}_{\text{TNNLS19}}$	33.4684 ↓5.45%	0.7571 ↓11.94%	0.9828 ↓0.54%	21.0599 ↑3.26%	1.5822 ↑18.22%	4.4666 ↑22.10%
	$\text{CSTF}_{\text{TCYB22}}$	33.1393 ↓6.38%	0.7645 ↓11.08%	0.9796 ↓0.86%	26.9870 ↑32.32%	1.6574 ↑23.84%	4.7199 ↑29.03%
	$\text{LTMR}_{\text{TIP22}}$	33.3507 ↓5.78%	0.7494 ↓12.84%	0.9823 ↓0.59%	20.7018 ↑1.50%	1.6004 ↑19.58%	4.5283 ↑23.79%
	$\text{CTRF}_{\text{PR22}}$	32.3756 ↓8.54%	0.7039 ↓18.13%	0.9722 ↓1.61%	25.6920 ↑25.97%	1.7804 ↑33.03%	5.3795 ↑47.06%
	$\text{LFCTNN}_{\text{TRGS24}}$	34.4480 ↓2.68%	0.8166 ↓5.02%	0.9843 ↓0.38%	26.5492 ↑30.17%	1.4569 ↑8.86%	4.0853 ↑11.68%
	$\text{SDP}_{\text{TRGS24}}$	33.2586 ↓6.04%	0.7785 ↓9.46%	0.9801 ↓0.81%	25.9541 ↑27.25%	1.6015 ↑19.67%	4.5145 ↑23.44%
	$\text{GTNN}_{\text{TNNLS25}}$	33.7165 ↓4.75%	0.7564 ↓12.03%	0.9873 ↓0.45%	20.3956 ↑0.00%	1.5337 ↑14.60%	4.3704 ↑19.48%
	$\text{LLTNN}_{\text{Ours}}$	35.3977 ↓0.00%	0.8598 ↓0.00%	0.9881 ↓0.00%	24.3676 ↑19.47%	1.3383 ↑0.00%	3.6580 ↑0.00%

interest and the corresponding error maps, LTTR, LTMR, and CTRF exhibit larger residuals and fail to faithfully reproduce spatial details along building edges; in contrast, the proposed approach more closely matches the ground truth on both bands.

3) CU Dataset: According to Table II, LLTNN delivers especially pronounced gains in reconstruction accuracy on CU; for example, it improves PSNR by nearly 3 dB over the second-best LFCTNN. Fig. 5 presents gray-scale reconstructions on the 50th and 90th bands together with the corresponding error maps. In gray-scale views, LFCTNN, GTNN, and LLTNN produce sharper HR-HSI appearances; the error maps further indicate that LLTNN attains the smallest residuals.

4) CAVE Dataset: For CAVE, metrics are computed over all 32 scenes, and their averages are reported in Table II. The proposed LLTNN shows additional advantages, achieving more than 0.8-dB higher PSNR than the second-best method. Pseudocolor reconstructions formed by bands 9, 15, and 31 are generated for three representative scenes: Chart and Stuffed Toy, Paints, and Face, as shown in Figs. 6–8. Using the zoomed-in region in Fig. 6 as an example, CTRF exhibits prominent red/green striping artifacts, whereas LTTR, CSTF, LTMR, and GTNN alleviate these artifacts but still leave noticeable red–green speckles. LFCTNN better preserves chromatic details yet remains suboptimal near edges. Overall, LLTNN restores fine details while maintaining spatial smoothness.

D. Results on Real Data

Since the real-world data do not provide pristine ground-truth reference images, full-reference metrics, such as PSNR and SSIM, cannot be used in this case. Therefore, we adopted three no-reference quality metrics, namely, metric Q [58], average gradient (AG) [59], and blind image quality assessment (BIQA) [60], to quantitatively assess the reconstruction performance. Specifically, Q is a gradient-based quality index derived from the singular values of the image gradient matrix, which reflects the overall structural clarity of the reconstructed image. AG measures the average local intensity variation and

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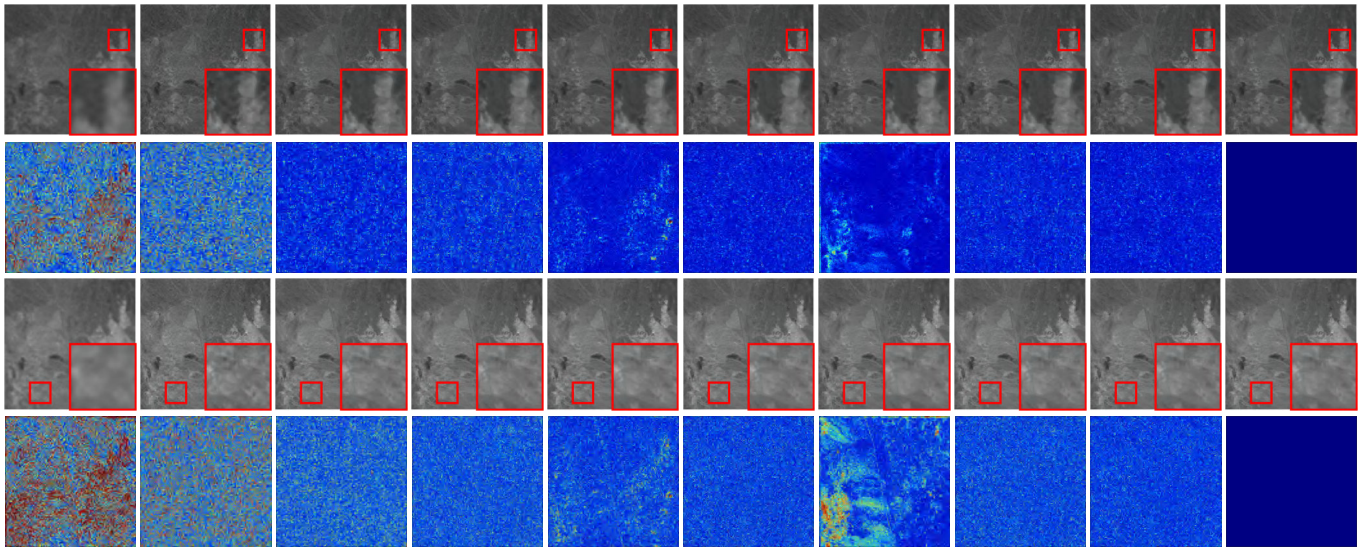


Fig. 5. Visual performance of completed approaches on the CU dataset: the top two rows, respectively, illustrate HSR results (band 50) and error images; the bottom two rows show the recovered HSIs (band 90) and corresponding error images.

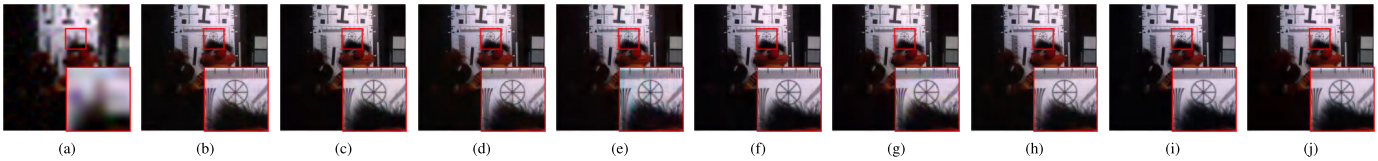


Fig. 6. Pseudocolor images (31st, 15th, and ninth bands) for Chart and Stuffed Toy (an MSI from the CAVE dataset). (a) LRHSI. (b) LTTR [51]. (c) CSTF [50]. (d) LTMR [18]. (e) CTRF [52]. (f) LFCTNN [22]. (g) SDP [34]. (h) GTNN [17]. (i) LLTNN. (j) GT.

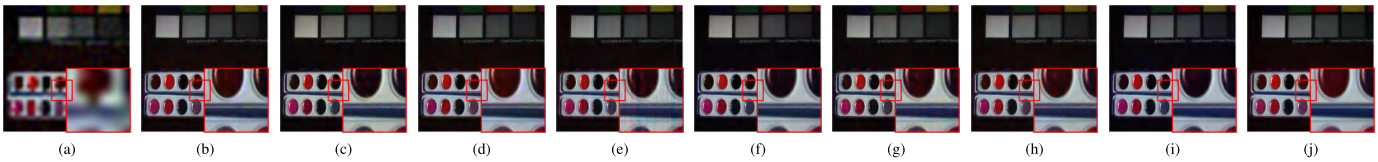


Fig. 7. Pseudocolor images (31st, 15th, and ninth bands) for Paints (an MSI from the CAVE dataset). (a) LRHSI. (b) LTTR [51]. (c) CSTF [50]. (d) LTMR [18]. (e) CTRF [52]. (f) LFCTNN [22]. (g) SDP [34]. (h) GTNN [17]. (i) LLTNN. (j) GT.

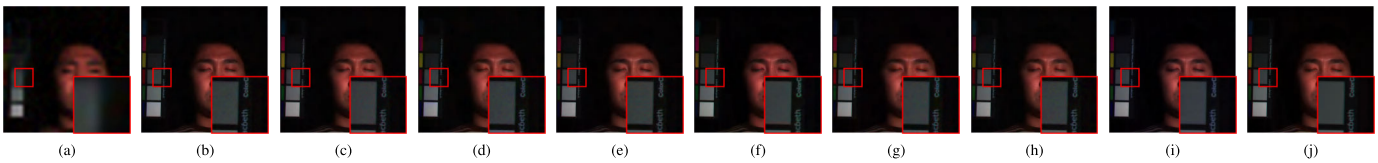


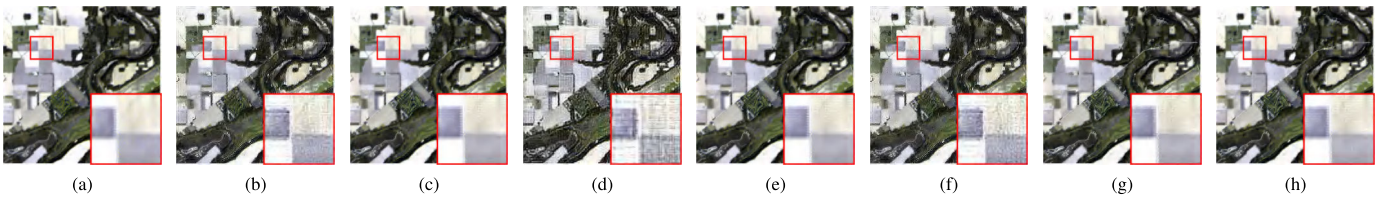
Fig. 8. Pseudocolor images (31st, 15th, and ninth bands) for Face (an MSI from the CAVE dataset). (a) LRHSI. (b) LTTR [51]. (c) CSTF [50]. (d) LTMR [18]. (e) CTRF [52]. (f) LFCTNN [22]. (g) SDP [34]. (h) GTNN [17]. (i) LLTNN. (j) GT.

is commonly used to characterize edge sharpness and detail richness. BIQA evaluates perceptual image quality without requiring a reference image. For all three metrics, larger values indicate better image quality.

As shown in Table III, the proposed LLTNN achieves the best performance on all three no-reference metrics, with Q = 0.15354, AG = 0.17282, and BIQA = 0.01865, which demonstrates its superior ability to preserve image structures, enhance spatial details, and improve perceptual quality on real-world data. Among the compared methods, CTRF is the most competitive baseline and achieves the second-best results across all three metrics, but it still remains consistently inferior to LLTNN. CSTF and SDP show moderate performance, indicating that they can recover part of the useful information but are still less effective in preserving fine details and perceptual quality. In contrast, GTNN, LTMR, LFCTNN, and LTTR obtain relatively lower scores, suggesting comparatively weaker reconstruction capability on the real-world dataset. Overall, these quantitative results are consistent with the visual comparisons and provide additional evidence for the effectiveness of the proposed method in practical real-data scenarios.

Fig. 9 presents the pseudocolor fused images (composited using bands 31, 15, and 9) generated by the seven competing approaches (a–g). The visual comparison clearly highlights the performance discrepancies among the models. The result from

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 9. Pseudocolor images (tenth, ninth, and first bands) from the real dataset. (a) LTTR [51]. (b) CSTF [50]. (c) LTMR [18]. (d) CTRF [52]. (e) LFCTNN [22]. (f) SDP [34]. (g) GTNN [17]. (h) LLTNN.

TABLE III
QUANTITATIVE COMPARISON RESULTS OF DIFFERENT METHODS ON THE REAL DATASET

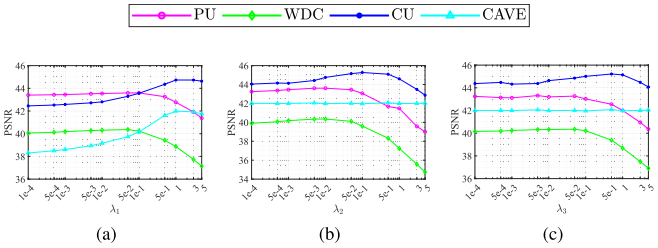
Method	Q	AG	BIQA
LTTR	0.10375	0.05304	0.01003
CSTF	0.11849	0.12189	0.01250
LTMR	0.09768	0.05928	0.00953
CTRF	0.14541	0.16281	0.01656
LFCTNN	0.09915	0.05889	0.00963
SDP	0.11668	0.12905	0.01253
GTNN	0.09235	0.06252	0.00952
LLTNN	**0.15354**	**0.17282**	**0.01865**

CTRF is the poorest, suffering from severe oversharpening and blocky effects that introduce prominent spatial artifacts and unnatural textures. Conversely, LTTR and CSTF exhibit the opposite problem, producing markedly blurry reconstructions with significant loss of spatial detail. LTMR achieves better sharpness, but its result remains visually "soft." LFCTNN, GTNN, and our proposed LLTNN significantly outperform all other methods, successfully reconstructing sharp spatial structures and edges without introducing noticeable artifacts. Among these top-tier results, our method achieves a high level of visual clarity while excelling in texture smoothness and color naturalness, confirming the effectiveness and robustness of the LLTNN model when applied to real-world data.

E. Discussion

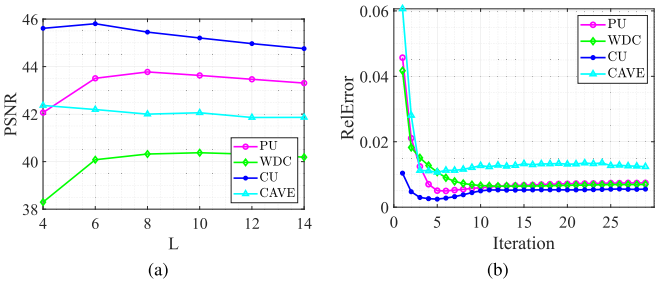
1) Parameter Analysis: The performance of LLTNN is influenced by two critical factors. First, as a low-rank tensor regularization method, the weights of the regularizers, λ_1, λ_2, and λ_3, directly affect performance. Second, low-dimensional subspace modeling, governed by dimension L, is a core premise for HSI-MSI fusion, as it dictates the balance between detail retention and noise suppression, while also impacting computational efficiency. We, therefore, analyze the sensitivity of λ_k and L on the PU, WDC, CU, and CAVE, using the *Thread spools* image datasets.

For the regularization parameters λ_k, Fig. 10(a)–(c) shows PSNR as a function of λ_1, λ_2, and λ_3. In our model, λ_1 and λ_2 mainly control the regularization strength associated with spatial structural priors, encouraging spatial consistency and low-rank correlations across spatial modes, while λ_3 primarily regulates the spectral constraint to capture the inherent spectral correlations of hyperspectral data. For PU (pink line), similar trends are observed: PSNR remains relatively stable within a moderate parameter range but declines rapidly as all λ_k's



(a) (b) (c)

Fig. 10. Impact of regularization parameters λ_1, λ_2, and λ_3 on PSNR. (a) λ_1. (b) λ_2. (c) λ_3.



(a) (b)

Fig. 11. Sensitivity and convergence analyses of the proposed method. (a) Impact of subspace dimension L on PSNR. (b) Convergence behavior in terms of RelError.

increase beyond 5e−2, indicating that overly strong regularization may oversmooth spatial or spectral structures. For WDC (green line), the performance peaks earlier, around $\lambda_k \approx$ 1e−2, before gradually decreasing. For CU (blue line), the PSNR reaches its maximum at $\lambda_1 = 1$, $\lambda_2 =$ 1e−1, and $\lambda_3 =$ 5e−1, suggesting that a balanced spatial and spectral regularization leads to the best reconstruction quality. The behavior on CAVE (cyan line) is slightly different: PSNR is most sensitive to λ_1 (increasing until $\lambda_1 = 1$), whereas varying λ_2 and λ_3 has relatively limited impact, which may be attributed to the relatively smooth spatial structures and stable spectral correlations in this dataset. As for the subspace dimension L, Fig. 11(a) indicates that PSNR reaches its maximum at $L = 8$ for PU and $L = 10$ for WDC. For CU and CAVE, the peak performance is achieved at $L = 6$. For all datasets, PSNR tends to decrease when L becomes too large (e.g., $L > 10$), suggesting that an overly large subspace dimension does not improve the reconstruction performance.

2) Convergence Analysis: The convergence of the proposed LLTNN is validated on PU, WDC, CU, and *Thread Spools* (CAVE). As shown in Fig. 11(b), the relative error (RelError) consistently decreases with the number of iterations and stabilizes at a steady value for all four cases, indicating that the algorithm has converged. While a strict mathematical proof of global convergence is challenging due to the dynamic updates of the implicit transform basis $\mathbf{U}^t$, we empirically

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TABLE IV

NUMERICAL RESULTS OF DIFFERENT HIERARCHICAL TRANSFORM COMBINATIONS ON THE PU DATASET WHEN $r = 4$ (BOLD: BEST; UNDERLINE: SECOND BEST). ABBREVIATIONS: L = LEARNABLE, F = DFT, C = DCT, AND T = TIGHT FRAMELET. HERE, THE NOTATION F∘C∘L DENOTES A PROGRESSIVE TRANSFORM SEQUENCE FROM LEFT TO RIGHT, I.E., DFT FOLLOWED BY DCT AND THEN THE LEARNABLE TRANSFORM

Case	L∘F	F∘L	L∘C	C∘L	L∘T	T∘L
PSNR	43.0448	43.3014	39.7372	41.0698	42.1767	42.1767
Case	L∘F∘C	L∘C∘F	F∘L∘C	F∘C∘L	C∘L∘F	C∘F∘L
PSNR	43.4595	43.3928	43.1346	**43.7814**	43.0305	<u>43.6045</u>
Case	L∘F∘T	L∘T∘F	F∘L∘T	F∘T∘L	T∘L∘F	T∘F∘L
PSNR	42.5605	42.5135	43.4058	43.4058	42.5219	43.4058
Case	L∘C∘T	L∘T∘C	C∘L∘T	C∘T∘L	T∘L∘C	T∘C∘L
PSNR	42.3653	42.3550	42.3769	42.3769	42.4047	42.3767

validate the stability of our ADMM algorithm. Fig. 11(b) plots the RelError versus the number of iterations across four simulated datasets. As observed, despite the per-iteration SVD updates, the RelError strictly and monotonically decreases, eventually converging to a stable plateau. This demonstrates that the intrinsic orthogonal constraint of the implicit basis ($\mathbf{U}^T\mathbf{U} = \mathbf{I}$), naturally guaranteed by the t-SVD framework, effectively bounds the search space and prevents optimization instability or overfitting to noise.

3) Transform-Combination Comparison: Different hierarchical transform combinations construct multidomain representations with distinct structural properties, which may affect the effectiveness of low-rank modeling for HSIs. To further verify the replaceability of the explicit transform module, we extend the original DFT-/DCT-based comparison by additionally introducing the tight framelet transform. The tight framelet transform can be viewed as a framelet/wavelet-type multiscale transform and has been used to define framelet-based TNN for third-order tensor recovery [61]. In Table IV, F, C, T, and L denote DFT, DCT, tight framelet, and learnable transform, respectively.

As reported in Table IV, the configuration F∘C∘L achieves the best performance with a PSNR of 43.7814 dB, while C ∘ F ∘ L obtains the second-best result with a PSNR of 43.6045 dB. This suggests that the DFT-DCT explicit transform used in the proposed model is an effective choice under the current experimental setting. Meanwhile, the framelet-related combinations also achieve competitive performance. For example, F ∘ L ∘ T, F ∘ T ∘ L, and T ∘ F ∘ L all obtain a PSNR of 43.4058 dB, outperforming several DFT/DCT-based alternatives, such as F ∘ L ∘ C and C ∘ L ∘ F. These results provide empirical evidence that the explicit transform module is not restricted to DFT or DCT but can also be extended to framelet-type transforms.

The superiority of the DFT → DCT sequence over DCT → DFT stems from the intrinsic hierarchical correlation structure of HSI.

1) *Global Spectral Decorrelation (DFT):* HSI spectral signatures typically possess strong long-range dependencies across the entire wavelength range. The DFT is employed as the primary stage to project these globally correlated spectral vectors into a sparse conjugate-symmetric frequency domain. This stage effectively decouples the complex spectral–spatial correlations by isolating the dominant spectral components in the complex plane.
2) *Local Spatial Energy Compaction (DCT):* Once spectral redundancy is addressed, the resulting frontal slices represent residual spatial structures. Given that spatial information is characterized by piecewise-smooth regions and localized edges, the DCT, which implicitly assumes symmetric boundary extensions, is more efficient than the DFT at compacting energy and suppressing boundary discontinuities (e.g., ringing artifacts).

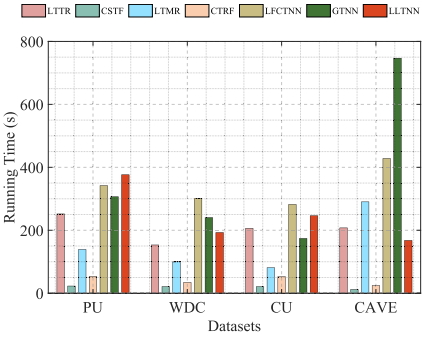


Fig. 12. Comparison of computational efficiency among different methods. The running time is measured in seconds.

This ordering is grounded in classical transform coding theory. The interband covariance of uniformly sampled HSI spectra approximates a Toeplitz structure, for which the DFT provides asymptotically optimal decorrelation [62]. After spectral decorrelation, the residual spatial content within each frequency slice is well characterized by a first-order Markov model, for which the DCT is asymptotically equivalent to the Karhunen–Loève Transform (KLT), i.e., the linear transform that maximizes energy compaction [63]. Consequently, the DFT→DCT cascade successively targets the two dominant correlation sources, spectral and spatial, in the order that maximizes the overall transform coding gain. From an optimization perspective, applying spectral DFT first provides a more homogeneous input for the spatial DCT, leading to higher total energy concentration in the transform domain. As shown in Table IV, reversing this order forces the DCT to handle highly correlated spectral data in its first stage, which limits the degrees of freedom for subsequent spatial learning and results in a 0.17-dB drop in PSNR.

Overall, the replacement experiment provides two observations. First, the explicit transform module is flexible and can be extended to alternative analytical bases, such as tight framelets. Second, the choice and ordering of explicit transforms still affect the final reconstruction accuracy.

4) Computational Efficiency: Fig. 12 plots the runtime of all methods across the datasets. It is evident that CSTF, CTRF, and LTMR are more computationally efficient, owing to their simpler model architectures. In contrast, LLTNN incorporates the core hierarchical transform module, which integrates MD

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TABLE V
COMPONENTWISE ABLATION STUDY ON THE PU DATASET

Variant	PSNR	SSIM	CC	SAM	ERGAS	DD
$\lambda_1 = 0$	43.5977	0.9865	0.9972	1.9702	1.0526	1.2419
$\lambda_2 = 0$	43.3717	0.9858	0.9969	2.0397	1.0703	1.2809
$\lambda_3 = 0$	43.6852	0.9870	0.9972	1.9396	1.0396	1.2236
w/o HPT	39.3927	0.9673	0.9916	2.8951	1.7246	2.0255
w/o Log	41.6193	0.9813	0.9951	2.3046	1.3353	1.5449
LLTNN	**43.7814**	**0.9874**	**0.9972**	**1.9065**	**1.0285**	**1.2041**

TABLE VI
PERFORMANCE COMPARISON OF SINGLE-, DUAL-, AND FULL-MODE REGULARIZATIONS

Method	PU	WDC	CU
single-mode	42.7724	38.6569	41.6909
dual-mode	43.4631	40.1396	42.6021
full-mode	**43.7834**	**40.4822**	**45.8075**

transforms with data-driven learnable transforms and performs SVD with full-mode regularization iteratively. This design prioritizes capturing complex spatial–spectral correlations over raw speed, resulting in a moderate runtime that is slower than the basic factorization models but remains competitive with, or often faster than, other SOTA methods, such as LFCTNN-FL and GTNN.

5) Ablation Study: To verify the effectiveness and necessity of each component, we conduct componentwise ablation studies on the PU dataset. Table V evaluates three key designs of LLTNN, including the full-mode LLTNN regularization, the HPT operator, and the log-sum-based nonconvex penalty. Specifically, $\lambda_1 = 0$, $\lambda_2 = 0$, and $\lambda_3 = 0$ are used as concise labels for the three modewise ablation variants, where the LLTNN regularization branch along the first, second, and third modes is removed, respectively. These variants are implemented by excluding the corresponding regularizer and its associated proximal update, rather than by simply setting the numerical value of λ_k to zero while keeping the full solver unchanged. The w/o HPT variant replaces the proposed HPT with a single DCT transform, aiming to examine the contribution of the explicit-to-implicit transform design. The w/o log variant removes the log-sum penalty and, thus, evaluates the contribution of the nonconvex singular-value regularization.

As shown in Table V, removing the mode-1 or mode-3 LLTNN causes only minor degradation, whereas ablating the mode-2 LLTNN yields a relatively larger performance drop. This indicates that different tensor modes contribute unequally to the reconstruction, and jointly regularizing all modes is beneficial for capturing the intrinsic spatial–spectral correlations of HSIs. Removing the HPT component also leads to clear degradation. Compared with the full LLTNN model, the w/o HPT variant decreases PSNR from 43.7814 to 39.3927 dB and SSIM from 0.9874 to 0.9673. Meanwhile, the spectral and global error metrics also deteriorate, with SAM increasing from 1.9065 to 2.8951 and ERGAS increasing from 1.0285 to 1.7246. These results indicate that the HPT component improves both structural reconstruction and spectral consistency under the tested setting. This improvement can be attributed to the explicit-to-implicit design: fixed explicit transforms provide stable structural constraints, while the implicit learnable transform adapts the representation to scene-specific correlations.

In addition, removing the log-sum-based regularizer also causes evident degradation. Compared with LLTNN, the w/o log variant decreases PSNR by 2.1621 dB and yields worse SAM, ERGAS, and DD values. This indicates that the nonconvex log-sum penalty is not redundant; it alleviates the overpenalization of dominant singular values and better preserves essential structural information in the transformed tensor representation. For readability, the detailed mathematical forms of these ablation variants are provided in the supplementary material.

To further verify the necessity of full-mode regularization, we compare the performance of single-mode, dual-mode, and full-mode constraints. Specifically, the single-mode and dual-mode settings impose the LLTNN regularizer on one and two tensor modes, respectively, while the full-mode setting jointly regularizes all three modes. As reported in Table VI, the full-mode regularization consistently achieves the best performance on all datasets. For example, on the PU dataset, the PSNR improves from 42.7724 (single-mode) and 43.4631 (dual-mode) to 43.7834 with full-mode regularization. Similar improvements can also be observed on the WDC and CU datasets. These results demonstrate that jointly exploiting the correlations across all tensor modes is beneficial for capturing the intrinsic low-rank structure of hyperspectral data, thereby leading to better restoration performance.

VII. CONCLUSION

This article proposed a novel HSI-SR framework based on the HPT and LLTNN regularizers. HPT resolves the rigidity–flexibility tradeoff by combining an explicit multidomain layer for robust structural representation with an implicit adaptive layer for scene-specific refinement. The LLTNN regularizer exploits low-rankness across all tensor modes with a nonconvex penalty that preserves essential features. Experiments on simulated and real-world datasets validate the superiority of our method over state-of-the-art approaches. Although the proposed LLTNN achieves promising reconstruction performance, the HPT and full-mode regularizations introduce additional computational cost compared with simpler single-transform models. In future work, we will explore more efficient optimization strategies to reduce the computational burden of the proposed framework. Moreover, since real-world HSIs usually exhibit complex nonlinear spatial–spectral dependencies, another promising direction is to investigate nonlinear transform mechanisms or nonlinear representation learning strategies as possible extensions of the current HPT framework. Such extensions may further enhance the ability of transform-domain low-rank models to characterize the intrinsic nonlinear structures of HSI data.

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